Attitude and Size Estimation of Satellite Targets Based on Imaging Geometry Analysis for Space Observation

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Attitude and size estimation of satellite targets is crucial for space surveillance. Existing methods that utilize radar or optical images from ground-based sensors typically rely on the basic imaging projection model. These methods require long-time or multistation observation to provide sufficient views and struggle to achieve accurate estimations under conditions of small view differences. To mitigate reliance on extensive views, we propose a method based on imaging geometry analysis for space observation. In our work, five crucial properties about the imaging geometry of rectangular solar panels are explored. Compared with the basic projection model, these properties impose additional constraints on the estimation process, thereby decreasing estimation errors by approximately 50% under limited observation conditions. Then, guided by these properties, we establish an over-determined system of equations to estimate two side vectors of the rectangle from their projections in images. To solve this complex system, the coarse estimation with a weighting strategy is conducted to decouple the system into linear subsystems, and these subsystems are solved by an Alternating Iterative Reweighted Least Square algorithm. The accuracy, real-time performance, and noise robustness of our method are demonstrated by experiments under various observation conditions.

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I. INTRODUCTION

With advancements in aerospace technology, outer space has emerged as a commanding height in international strategic competition. Space situation awareness (SSA) has become a prominent research area due to its pivotal role in space security and control [1], [2], [3]. On-orbit state estimation of satellite targets is one of the core technologies within SSA. Estimated parameters, such as attitude and size, provide important information for monitoring the health of space assets and analyzing the intention of noncooperative satellites [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. Ground-based radars or optoelectronic telescopes serve as primary observation sources. The observed inverse synthetic aperture radar (ISAR) and optical images can offer rich structural information about the targets. Absolute attitude and three-dimensional (3D) size of rectangular solar panels are typical features describing the satellite's on-orbit state, and numerous studies have focused on their estimation [4], [5], [8], [9], [10].

The existing methods for satellite attitude and size estimation can be divided into three basic categories. The first category is based on the template matching [16], [19], [20], which is only applicable to cooperative satellites. A complete template library must be established in advance using full-view images of the target, and the attitude estimation is realized by matching the features of measured images to those of templates. The second category involves deep-learning techniques [12], [13], [14]. These methods convert the estimation task into a classification problem by discretizing the attitude space of the target, and the nonlinear mapping between images and attitude space is learned using abundant labeled target images. In addition, these methods do not consider the imaging geometry and only estimate relative attitudes. The thirs category formulates the estimation as optimization problems based on the ISAR or optical imaging projection model [4], [5], [6], [7], [8], [9], [10], [15], [16], [17], [18], [21]. Regarding solar panels, the method in [4] constructs the optimization using the projection model of two rectangular sides, incorporating side orthogonality. The method in [8] builds the optimization based on the relationship between 3D parameters of a rectangle and principal component analysis (PCA) results of its 2D projections. These methods can estimate the absolute attitude of noncooperative targets. However, only partial imaging geometry properties in space observation are leveraged by these methods. For example, they do not analyze the geometric relationship between normal vectors of imaging planes and two rectangle sides, which could impose additional constraints on the feasible region of the optimization problems [4], [5], [8], [9], [10]. The limited utilization of imaging geometry properties results in their poor estimation accuracy when view differences are small. What's more, the optimizations in these methods are usually high-dimensional, nonconvex, and complex, making it challenging for optimization algorithms to achieve stable and rapid convergence toward the globally optimal solution [8]. Conclusively, their practical application is limited by the requirements of long-time observation and time-consuming optimization processes.

To promote practical applications, several researchers have introduced innovative approaches tailored to short-time observation scenarios [9], [10], [15]. The method in [15] takes advantage of the complementarity and correlation between optical and radar imaging. This method achieves accurate estimation with a pair of optical and radar images. However, the images should be observed simultaneously from the same angle, which incurs high engineering costs. The methods in [9] and [10] estimate solar panel parameters by analyzing the quadratic spatial-variant phases (QSVPs) caused by target rotation. Theoretically, these methods only need a single radar image with phase information for accurate estimation. However, these methods are not suitable for optical images, and the sensitive quadratic phases render them ineffective for ISAR images with low signal-to-noise (SNR) ratios [10]. These methods put forward additional requirements for the deployment or capability of sensors, and the imaging geometry properties in space observation remain underexplored.

To further enhance estimation accuracy under short-time observation with a single sensor, that is, when view differences are small, we propose a method based on imaging geometry analysis for space observation. In our work, five important properties about the imaging geometry of rectangular solar panels are unveiled and validated. These properties extend the basic imaging projection model utilized in the existing methods, providing additional information that alleviates the requirements for extensive views. Leveraging these properties, we establish an over-determined system of equations to estimate the side vectors of the solar panel from its image projections. However, the intertwined side vectors in this system impede deriving the optimal solution. We solve this complex system through the following three steps. First, linear systems of equations are established based on the basic imaging projection model to conduct the coarse estimation, and the validated properties constrain the coarse estimation process with a weighting strategy (WS). Second, by substituting one side vector with the rough initial solution, we decouple the complex system into linear subsystems—one for each side. Finally, these subsystems are alternatively and iteratively solved to achieve the final delicate estimation. To demonstrate the superiority of our method, comparative experiments with the existing methods proposed in [4], [8], and [10] are carried out. In addition, we analyze the noise-robustness and applicability to multimode data by conducting experiments on both radar and optical images with different SNRs. Our main contributions are listed as follows.

1) We propose a satellite attitude and size estimation method based on imaging geometry analysis. By exploring five crucial properties about imaging geometry in space observation, we establish a multiproperty constrained over-determined system of equations. Compared to existing methods, our method imposes stronger constraints on the estimation process and achieves higher estimation accuracies, especially when view differences are small.
2) To address the intertwined side vectors in the complex system, we derive a rough initial solution to decouple the system into linear subsystems—one for each side. Then, an Alternating Iterative Reweighted Least Square (AIRLS) algorithm is designed to solve these subsystems.
3) To ensure the accuracy of the initial solution without introducing intertwined side vectors into equations, we adopt a WS to impose the multiproperty constraints on the coarse estimation.

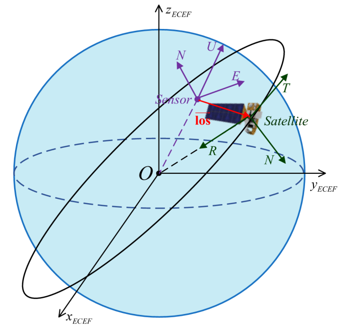

Fig. 1. Diagram of the observation geometry. The axes of the ENU system are marked in purple. The axes of the RTN system are marked in green. The LOS vector is marked in red.

The rest of this article is structured as follows. In Section II, the basic imaging geometry in the SSA scenario is introduced. In Section III, we further explore five properties of the imaging geometry. In Section IV, the estimation method based on imaging geometry analysis is described in detail. In Section V, several experiments are conducted to verify the superiority of our method. Section VI concludes the article.

II. Basic Imaging Geometry in SSA Scenario

The observation geometry between ground-based sensors and satellites in orbit is shown in Fig. 1. During the observation, the attitude of a three-axis stabilized satellite can be considered unchanged in a local orbital coordinate system called RTN coordinate system [22], whose axes are defined by radial direction (R), along-track direction (T), and cross-track direction (N). The observation station is located at the origin of a site-centric (East-North-Up, ENU) coordinate system. The instantaneous line of sight (LOS) is provided by the tracking information of the sensor, and the LOSs in the RTN system can be further obtained by coordinate transformation

$$\overrightarrow{\mathbf{los}}_{\mathrm{RTN},t} = \mathbf{R}_{\mathrm{ENU}}^{\mathrm{RTN}} \overrightarrow{\mathbf{los}}_{\mathrm{ENU},t} \tag{1}$$

where $\overrightarrow{\mathbf{los}}_{\mathrm{ENU},t}, \overrightarrow{\mathbf{los}}_{\mathrm{RTN},t} \in \mathbb{R}^{3\times1}$ are, respectively, the LOS vectors in the ENU and RTN systems at time t, $\mathbf{R}_{\mathrm{ENU}}^{\mathrm{RTN}} \in \mathbb{R}^{3\times3}$ is the rotation matrix determined based on the Kepler two-body motion property [22].

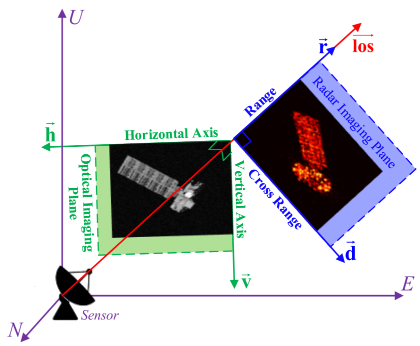

Fig. 2. Diagram of the imaging projection model for optical and radar systems.

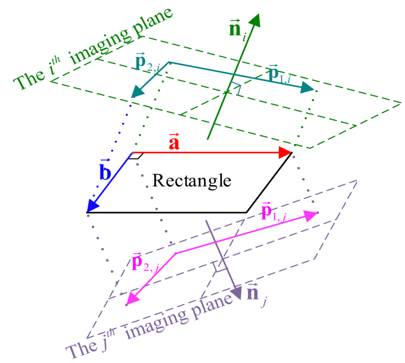

Fig. 3. Imaging geometry of a planar rectangle in 3D space.

In [17], the imaging process is described as projecting the target from the 3D place to a 2D imaging plane, and the basic imaging projection model is described as

$$\mathbf{H}\overrightarrow{\mathbf{tp}} = \begin{bmatrix} px \cdot \Delta d_x \\ py \cdot \Delta d_y \end{bmatrix} \tag{2}$$

where $\overrightarrow{\mathbf{tp}} \in \mathbb{R}^{3\times 1}$ is the 3D location of a target point in the RTN system, $[px, py]^T$ is the location of its projection pixel; Δd_x and Δd_y are the image resolutions, whose units are "meters/pixel." $\mathbf{H} \in \mathbb{R}^{2\times 3}$represents the imaging projection matrix. As shown in Fig. 2, the imaging planes for optical and radar systems are different due to their unique imaging mechanisms.

Optical imaging relies on the affine phenomenon, and its imaging plane is determined by the horizontal and vertical axes, which are perpendicular to the LOS [17]. In our work, the horizontal axis $\vec{\mathbf{h}}$ is defined as horizontal to the ground surface

$$\vec{\mathbf{h}} = \mathbf{R}_{\text{ENU}}^{\text{RTN}} \left(\overrightarrow{\mathbf{los}}_{\text{ENU},t_I} \otimes \vec{\mathbf{g}} \right) \tag{3}$$

where t_I is the imaging moment, $\vec{\mathbf{g}} \in \mathbb{R}^{3\times 1}$ is a vector from the Earth's center to the observation station, $\otimes$ is the cross product; The vertical axis $\vec{\mathbf{v}}$ is determined according to the Right-hand Rule

$$\vec{\mathbf{v}} = \overrightarrow{\mathbf{los}}_{\text{RTN},t_I} \otimes \vec{\mathbf{h}}. \tag{4}$$

Then, the imaging projection matrix of the optical system is represented as

$$\mathbf{H} = \begin{bmatrix} \vec{\mathbf{v}} & \vec{\mathbf{h}} \end{bmatrix}^T. \tag{5}$$

The resolution of optical images can be obtained by the optical calibration methods [35].

Radar imaging records the target distance history as one dimension of the image, and the target characteristics in the other dimension are mined from the frequency domain of signals by coherent processing on multiple continuous radar echoes. Therefore, its imaging plane is determined by the range and doppler axes [23]. The range axis $\vec{\mathbf{r}}$ is aligned with the LOS at the central moment t_c of the coherent processing interval (CPI)

$$\vec{\mathbf{r}} = \overrightarrow{\mathbf{los}}_{RTN,t_c}. \tag{6}$$

The doppler axis $\vec{\mathbf{d}}$ is defined as the derivative of the LOSs during the CPI

$$\vec{\mathbf{d}} = \overrightarrow{\mathbf{de}} / \left\| \overrightarrow{\mathbf{de}} \right\|_2, \ \overrightarrow{\mathbf{de}} = \frac{\partial \overrightarrow{\mathbf{los}}_{\text{RTN},t}}{\partial t} \Big|_{t=t_c}. \tag{7}$$

Then, the imaging projection matrix of the radar system is represented as

$$\mathbf{H} = \begin{bmatrix} \vec{\mathbf{r}} \, \vec{\mathbf{d}} \end{bmatrix}^T. \tag{8}$$

Its range and cross-range resolutions are calculated as

$$\Delta d_x = \Delta d_r = \frac{c}{2B}, \Delta d_y = \Delta d_d = \frac{\lambda}{2\Omega} \tag{9}$$

where c is the light speed, B is the signal bandwidth, λ is the signal wavelength, and Ω is the relative rotation angle during the CPI, which can be estimated by the method in [29].

III. FIVE PROPERTIES ABOUT IMAGING GEOMETRY IN SPACE OBSERVATION

Based on the basic imaging geometry [17], [23], we further explore the related properties for rectangular solar panels. In this section, five important properties are unveiled, and the proofs are given in Algorithm 1.

As shown in Fig. 3, a planar rectangle in 3D space can be described by its two orthogonal adjacent side vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$, $\vec{\mathbf{a}}, \vec{\mathbf{b}} \in \mathbb{R}^{3\times 1}$, $\vec{\mathbf{a}}^T\vec{\mathbf{b}} = 0$, $\|\vec{\mathbf{a}}\|_2 = a$, and $\|\vec{\mathbf{b}}\|_2 = b$. The normal vector of the rectangle is defined as

$$\vec{\mathbf{z}} = (\vec{\mathbf{a}}/a) \otimes \left(\vec{\mathbf{b}}/b\right). \tag{10}$$

Two imaging planes (ith and jth, $i, j \in \mathrm{N}^+$, $i \neq j$) are determined by their normal vectors $\vec{\mathbf{n}}_i$ and $\vec{\mathbf{n}}_j$. These normal vectors can be represented as

$$\vec{\mathbf{n}}_i = x_i \cdot \left(\frac{\vec{\mathbf{a}}}{a}\right) + y_i \cdot \left(\frac{\vec{\mathbf{b}}}{b}\right) + z_i \cdot \vec{\mathbf{z}} \tag{11}$$

$$\vec{\mathbf{n}}_j = x_j \cdot \left(\frac{\vec{\mathbf{a}}}{a}\right) + y_j \cdot \left(\frac{\vec{\mathbf{b}}}{b}\right) + z_j \cdot \vec{\mathbf{z}} \tag{12}$$

TABLE I
Notations of Some Vectors and Planes

$\vec{\mathbf{p}}_{k,i} \in \mathbb{R}^{3\times1}$	The k^{th} projection component in the i^{th} image
$\vec{\mathbf{p}}_{k,j} \in \mathbb{R}^{3\times1}$	The k^{th} projection component in the j^{th} image
$Pl_{k,i}$	Plane constructed by $\vec{\mathbf{p}}_{k,i}$ and $\vec{\mathbf{n}}_i$
$Pl_{k,j}$	Plane constructed by $\vec{\mathbf{p}}_{k,j}$ and $\vec{\mathbf{n}}_j$
$\vec{\mathbf{l}}_{1,i,j} \in \mathbb{R}^{3\times1}$	Direction of intersection line between $Pl_{1,i}$ and $Pl_{1,j}$
$\vec{\mathbf{l}}_{2,i,j} \in \mathbb{R}^{3\times1}$	Direction of intersection line between $Pl_{2,i}$ and $Pl_{2,j}$
$\vec{\mathbf{l}}_{3,i,j} \in \mathbb{R}^{3\times1}$	Direction of intersection line between $Pl_{1,i}$ and $Pl_{2,j}$
$\vec{\mathbf{l}}_{4,i,j} \in \mathbb{R}^{3\times1}$	Direction of intersection line between $Pl_{2,i}$ and $Pl_{1,j}$
$\vec{\mathbf{n}}_{z,i,j} \in \mathbb{R}^{3\times1}$	Cross product of $\vec{\mathbf{n}}_i$ and $\vec{\mathbf{n}}_j$

where $x_i^2 + y_i^2 + z_i^2 = 1$ and $x_j^2 + y_j^2 + z_j^2 = 1$. The normal vectors of imaging planes should satisfy the following two assumptions to make the estimation feasible.

ASSUMPTION 1 $\vec{\mathbf{n}}_i^T \vec{\mathbf{z}} \neq 0$ and $\vec{\mathbf{n}}_j^T \vec{\mathbf{z}} \neq 0$(i.e., $z_i \neq 0, z_j \neq 0$).

When $\vec{\mathbf{n}}_i^T \vec{\mathbf{z}} = 0$, the projection of the rectangle in the ith image is a line rather than a parallelogram, and these images cannot be used in estimations [4].

ASSUMPTION 2 $\vec{\mathbf{n}}_i \neq \vec{\mathbf{n}}_j$; otherwise, projection components in these two images are identical.

Notations of some related vectors and planes are listed in Table I.

In Table I, $k = 1, 2$. We assume that $\vec{\mathbf{p}}_{1,i}$ and $\vec{\mathbf{p}}_{1,j}$ are projection components of $\vec{\mathbf{a}}$; $\vec{\mathbf{p}}_{2,i}$ and $\vec{\mathbf{p}}_{2,j}$ are projection components of $\vec{\mathbf{b}}$. This assumption can be validated through projection association in Section IV-B. Then, we explore five important properties about the geometric relationships between these vectors and plane.

PROPERTY 1 ("*Direction-consistency*"):
$\vec{\mathbf{l}}_{1,i,j}$ is parallel to $\vec{\mathbf{a}}$, i.e., $\vec{\mathbf{l}}_{1,i,j} = \vec{\mathbf{a}}/a$.
$\vec{\mathbf{l}}_{2,i,j}$ is parallel to $\vec{\mathbf{b}}$, i.e., $\vec{\mathbf{l}}_{2,i,j} = \vec{\mathbf{b}}/b$.

PROPERTY 2 ("*Orthogonality*"):
$\vec{\mathbf{l}}_{1,i,j}$ is always perpendicular to $\vec{\mathbf{l}}_{2,i,j} = \vec{\mathbf{b}}/b$
($\vec{\mathbf{l}}_{1,i,j}^T \vec{\mathbf{l}}_{2,i,j} = 0$, $PE_{2,1,i,j}$).
$\vec{\mathbf{l}}_{3,i,j}$ is not perpendicular to $\vec{\mathbf{l}}_{4,i,j}$ when $\vec{\mathbf{n}}_i^T \vec{\mathbf{n}}_j \neq 0$.
($\vec{\mathbf{l}}_{3,i,j}^T \vec{\mathbf{l}}_{4,i,j} \neq 0$, $PE_{2,2,i,j}$).

PROPERTY 3 ("*Projection-consistency*"):
$\vec{\mathbf{p}}_{1,i}^T \vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{p}}_{1,j}^T \vec{\mathbf{n}}_{z,i,j}$ and $\vec{\mathbf{p}}_{2,i}^T \vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{p}}_{2,j}^T \vec{\mathbf{n}}_{z,i,j}$ ($PE_{3,1,i,j}$ and $PE_{3,2,i,j}$) are always valid.
$\vec{\mathbf{p}}_{1,i}^T \vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{p}}_{2,j}^T \vec{\mathbf{n}}_{z,i,j}$ and $\vec{\mathbf{p}}_{2,i}^T \vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{p}}_{1,j}^T \vec{\mathbf{n}}_{z,i,j}$ ($PE_{3,3,i,j}$ and $PE_{3,4,i,j}$) are not valid when $\xi_a \cdot b \neq \xi_b \cdot a$. ($\xi_a = x_j z_i - x_i z_j$, $\xi_b = y_i z_j - y_j z_i$)

PROPERTY 4 ("*Ambiguity*"):
When $\xi_a \cdot b = \xi_b \cdot a$ and $\vec{\mathbf{n}}_i^T \vec{\mathbf{n}}_j = 0$, there is more than one planar rectangle, whose projection components on the imaging planes are $\vec{\mathbf{p}}_{k,i}$ and $\vec{\mathbf{p}}_{k,j}$ ($k = 1, 2$).

PROPERTY 5 ("*Ratio-equality*"):

$$\frac{\left(\vec{\mathbf{l}}_{1,i,j}^T \vec{\mathbf{l}}_{3,i,j}\right)\cdot\left(\vec{\mathbf{n}}_i^T \vec{\mathbf{l}}_{2,i,j}\right)}{\vec{\mathbf{n}}_i^T \vec{\mathbf{z}}} = \frac{\left(\vec{\mathbf{l}}_{2,i,j}^T \vec{\mathbf{l}}_{3,i,j}\right)\cdot\left(\vec{\mathbf{n}}_j^T \vec{\mathbf{l}}_{1,i,j}\right)}{\vec{\mathbf{n}}_j^T \vec{\mathbf{z}}} (PE_{5,1,i,j}).$$

$$\frac{\left(\vec{\mathbf{l}}_{1,i,j}^T \vec{\mathbf{l}}_{4,i,j}\right)\cdot\left(\vec{\mathbf{n}}_j^T \vec{\mathbf{l}}_{2,i,j}\right)}{\vec{\mathbf{n}}_j^T \vec{\mathbf{z}}} = \frac{\left(\vec{\mathbf{l}}_{2,i,j}^T \vec{\mathbf{l}}_{4,i,j}\right)\cdot\left(\vec{\mathbf{n}}_i^T \vec{\mathbf{l}}_{1,i,j}\right)}{\vec{\mathbf{n}}_i^T \vec{\mathbf{z}}} (PE_{5,2,i,j}).$$

For convenience, the hth equation in the mth property determined by the ith and jth image is abbreviated as $PE_{m,h,i,j}$. $PE^L_{m,h,i,j}$, and $PE^R_{m,h,i,j}$ respectively represent the left and right side of this equation.

To facilitate reader understanding, the diagrams for these properties are presented in Fig. 4.

In practice, the estimation process of the rectangle can be understood as inferring the rectangle sides $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ from the projection components $\vec{\mathbf{p}}_{k,i}$ and $\vec{\mathbf{p}}_{k,j}$. Under our assumption, $\vec{\mathbf{p}}_{1,i}$ and $\vec{\mathbf{p}}_{1,j}$ are projection components of the same side. Then, we can determine 3D directions of the rectangle sides by using $\vec{\mathbf{l}}_{1,i,j}$ and $\vec{\mathbf{l}}_{2,i,j}$ in accordance with Property 1. On the contrary, the directions should be determined by using $\vec{\mathbf{l}}_{3,i,j}$ and $\vec{\mathbf{l}}_{4,i,j}$ when $\vec{\mathbf{p}}_{1,i}$ and $\vec{\mathbf{p}}_{2,j}$ correspond to the same side. Therefore, it is inevitable to judge whether $\vec{\mathbf{p}}_{1,i}$ is associated with $\vec{\mathbf{p}}_{1,j}$ or with $\vec{\mathbf{p}}_{2,j}$. **Properties 2 and 3** indicate the differences between these vectors, which can serve as bases for judgment. Property 4 points out the condition for the tenability of our judgement ($\xi_a \cdot b \neq \xi_b \cdot a$ or $\vec{\mathbf{n}}_i^T \vec{\mathbf{n}}_j \neq 0$).

IV. ATTITUDE AND SIZE ESTIMATION BASED ON IMAGING GEOMETRY ANALYSIS

A. Method Flowchart

Our estimation method based on imaging geometry analysis is illustrated in Fig. 5. First, side projections of the solar panel are extracted and associated across images. Then, an over-determined system of equations is established guided by the proved geometric properties. To solve this complex system, an initial solution is provided by the coarse estimation to decouple the complex system into linear subsystems, and these subsystems are solved by the AIRLS algorithm to achieve the final delicate estimation. Details of each step are outlined in the subsequent.

Besides the vectors listed in Table I, some other notations are introduced in this section, which are summarized in Table II.

B. Extract and Associate Projection Components

1) *Extract Side Projections:* The 3D attitude and size of the rectangular solar panel can be determined by its two adjacent side vectors. As shown in Fig. 6, to extract projections of the sides from images, solar panels should be segmented. The segmentation can be realized by some manual operations or semantic segmentation methods [11]. Then, the side projections $\overrightarrow{\mathbf{Ip}}_{k,i} \in \mathbb{R}^{2\times1}$ in the ith image ($k = 1, 2$, $i = 1, 2, \ldots, N$, N is the number of

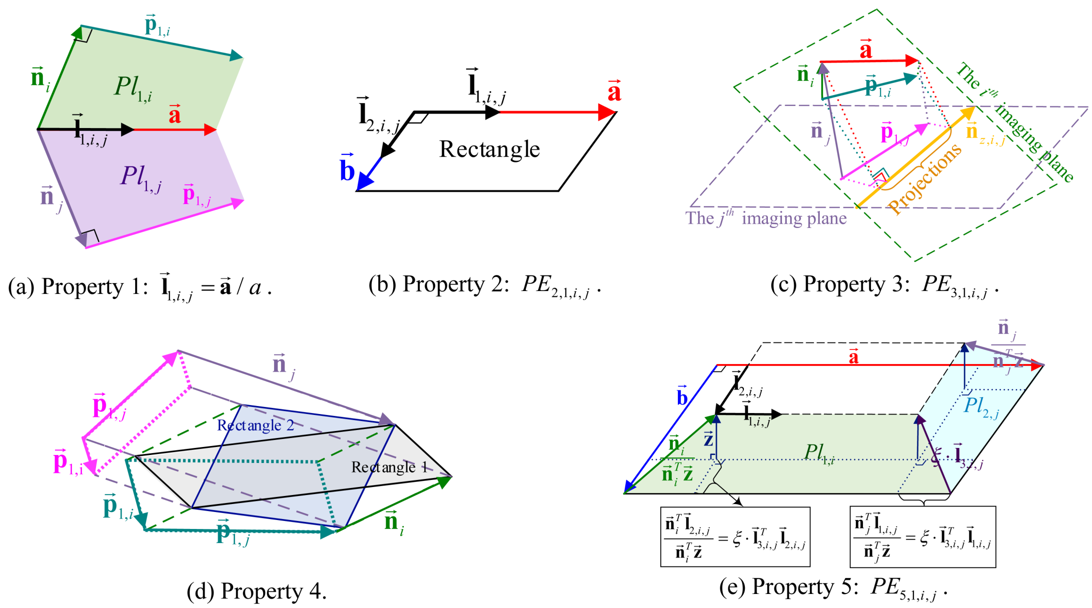

Fig. 4. Diagrams for five properties about imaging geometry. (a) Property 1: $\vec{\mathbf{l}}_{1,i,j} = \vec{\mathbf{a}}/a$. (b) Property 2: $PE_{2,1,i,j}$. (c) Property 3: $PE_{3,1,i,j}$. (d) Property 4. (e) Property 5: $PE_{5,1,i,j}$.

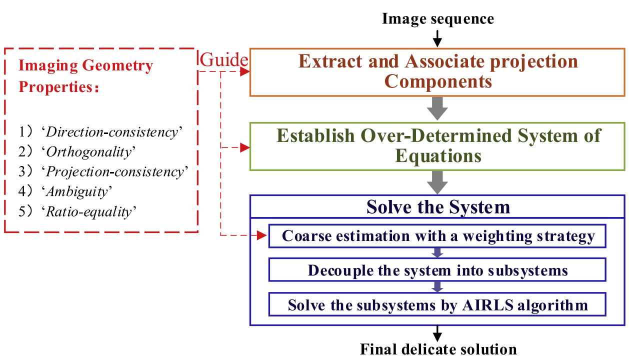

Fig. 5. Flowchart of our satellite attitude and size estimation method.

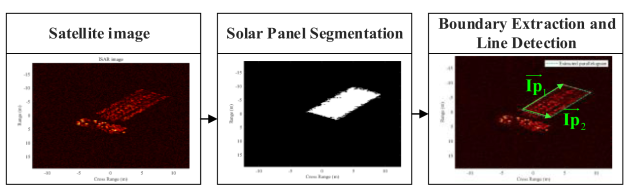

Fig. 6. Diagram of side projection extraction.

images) are extracted by boundary extraction and line detection algorithms [6]. These projections are scaled using image resolutions Δd_x and Δd_y. Then, their units are "meters."

The projection components $\vec{\mathbf{p}}_{k,i}$ mentioned in Section III can be further calculated as

$$\vec{\mathbf{p}}_{k,i} = \mathbf{H}_i^T \overrightarrow{\mathbf{Ip}}_{k,i} \tag{13}$$

where $\mathbf{H}_i$ is imaging projection matrix of the ith image.

TABLE II
Notations Introduced in Section IV

N	The number of images
$\overrightarrow{\mathbf{Ip}}_{k,i} \in \mathbb{R}^{2\times1}$	The k^{th} side projection in the i^{th} image
$\vec{\mathbf{p}}_{k,i} \in \mathbb{R}^{3\times1}$	The projection component corresponding to $\overrightarrow{\mathbf{Ip}}_{k,i}$
$\mathbf{H}_i$	The projection matrix for the i^{th} image
$J_{i,j}$	The judgement operator for associating projection components in the i^{th} and j^{th} images
$\mathcal{M}_i$	The suitability measurement operator for the i^{th} image
ζ_i	The weight coefficient for the i^{th} image
$\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$	The side vectors which need to be solved
$\vec{\mathbf{z}}$	The auxiliary vector
$\vec{\mathbf{e}}_{1,0}$, $\vec{\mathbf{e}}_{2,0}$, $\vec{\mathbf{z}}_0$	The initial solution for $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$, $\vec{\mathbf{z}}$
$\mathbb{S}_{Comp,L} = \mathbb{S}_{Comp,R}$	The established complex system of equations
$\mathbf{H}_A\vec{\mathbf{a}} = \vec{\mathbf{v}}_A$	The decoupled system for $\vec{\mathbf{a}}$
$\mathbf{H}_B\vec{\mathbf{b}} = \vec{\mathbf{v}}_B$	The decoupled system for $\vec{\mathbf{b}}$
$\mathbf{H}_Z\vec{\mathbf{z}} = \vec{\mathbf{v}}_Z$	The decoupled system for $\vec{\mathbf{z}}$
$\overrightarrow{\mathbf{rs}}$	The residual vector of equations
$\vec{\boldsymbol{\omega}}$, $\boldsymbol{\Psi}$	The weight vector and matrix for equations
Δe	The change of side vectors during iterations
$\mathcal{D}(a,b)$	The normalized residual function
$[\![f(i)]\!]_i$	The function of vertical arrangement
$\mathbf{A} \backslash \mathbf{B}$	The left division for matrixes
$\mathrm{diag}(\cdot)$	The diagonal function

2) *Associate Projection Components:* As discussed in Section III, the association of the projection components, i.e., the judgement, is the premise of the estimation. To associate the projection components $\vec{\mathbf{p}}_{k,i}$ and $\vec{\mathbf{p}}_{k,j}$ $(k = 1, 2)$,

which are respectively extracted from the ith and jth image, a judgement operator $J_{i,j}$ is defined as

$$\boldsymbol{J}_{i,j} = \boldsymbol{J}_{i,j,1} + \beta_1 \cdot \boldsymbol{J}_{i,j,2}. \tag{14}$$

$$\boldsymbol{J}_{i,j,1} = \left|PE^{L}_{2,1,i,j}\right| - \left|PE^{L}_{2,2,i,j}\right|. \tag{15}$$

$$\boldsymbol{J}_{i,j,2} = \sum_{h=1}^{2} \begin{pmatrix} \mathcal{D}\left(PE^{L}_{3,h,i,j}, PE^{R}_{3,h,i,j}\right) \\ -\mathcal{D}\left(PE^{L}_{3,h+2,i,j}, PE^{R}_{3,h+2,i,j}\right) \end{pmatrix}. \tag{16}$$

where $\boldsymbol{J}_{i,j,1}$ and $\boldsymbol{J}_{i,j,2}$ are, respectively, designed according to **Properties 2 and 3**; $\beta_1 \in (0, 1)$ is a constant coefficient, $\mathcal{D}(a, b)$ is a difference normalization function defined as (17). With the determined imaging projection matrices and extracted projection components, the vectors $\vec{\mathbf{n}}_i$, $\vec{\mathbf{n}}_j$, $\vec{\mathbf{n}}_{z,i,j}$, and $\vec{\mathbf{l}}_{1/2/3/4,i,j}$ in 'PE's can be calculated using space analytic geometry techniques, whose details are listed in Algorithm 2

$$\mathcal{D}(a, b) = \frac{|a - b|}{\max(|a|, |b|) + 10^{-5}}. \tag{17}$$

Based on this operator, the association between two images is realized as follows:

$$\begin{cases} \vec{\mathbf{p}}_{1,i} \Leftrightarrow \vec{\mathbf{p}}_{1,j}, \vec{\mathbf{p}}_{2,i} \Leftrightarrow \vec{\mathbf{p}}_{2,j} & \boldsymbol{J}_{i,j} < -\mathrm{Thr}_a \\ \vec{\mathbf{p}}_{1,i} \Leftrightarrow \vec{\mathbf{p}}_{2,j}, \vec{\mathbf{p}}_{2,i} \Leftrightarrow \vec{\mathbf{p}}_{1,j} & \boldsymbol{J}_{i,j} > \mathrm{Thr}_a \end{cases} \tag{18}$$

where $\Leftrightarrow$ represents the association relationship, $\mathrm{Thr}_a = 10^{-2}$ is a manual threshold. According to Property 4, the reliability of the judgement is too low when $|\boldsymbol{J}_{i,j}| < \mathrm{Thr}_a$, and the association should be further determined with other images.

When more than two images are available, a comprehensive decision should be further implemented. As shown in Algorithm 1, the comprehensive decision is realized by *Preliminary Association* and *Rechecking*.

1) In the *Preliminary Association*, each image is associated in turn with all previous images using the comprehensive judgement (CJ) operator $\boldsymbol{cJ}_j$

$$\boldsymbol{cJ}_j = \sum_{i=1}^{j-1} \boldsymbol{J}_{i,j}. \tag{19}$$

For the jth image, $\boldsymbol{J}_{i,j}$ $(i = 1, 2, \ldots, j-1)$ are its judgement operators with the previous images. According to (18), its initial association relationships with previous images are deemed correct when $\boldsymbol{cJ}_j \leq 0$, otherwise the values of $\vec{\mathbf{p}}_{1,j}$ and $\vec{\mathbf{p}}_{2,j}$ need be swapped.

2) In the *Rechecking*, we check whether the image is correctly associated with both previous and subsequent images. The rechecking judgement (RJ) operator $\boldsymbol{rJ}_j$ is defined as.

$$\boldsymbol{rJ}_j = \sum_{i \neq j} \boldsymbol{J}_{i,j}. \tag{20}$$

According to (18), its current association relationships with all the other images are correct when $\boldsymbol{rJ}_j \leq 0$, otherwise the values of $\vec{\mathbf{p}}_{1,j}$ and $\vec{\mathbf{p}}_{2,j}$ need be swapped.

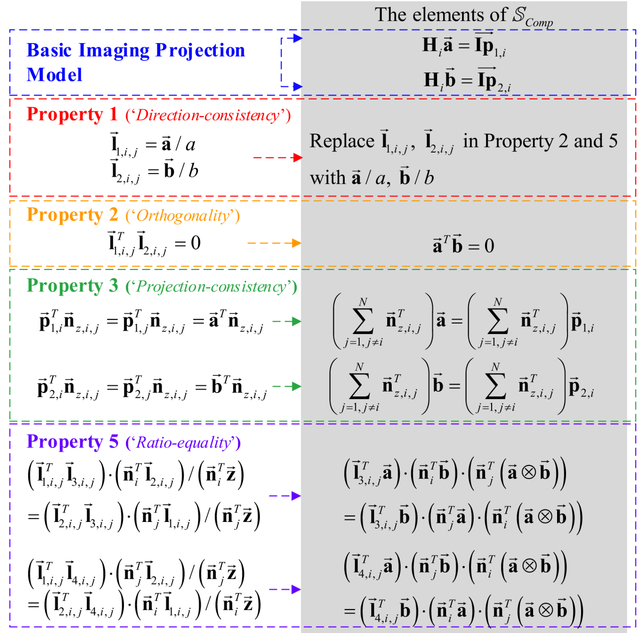

Fig. 7. Diagram of the established over-determined system. The left half gives the utilized properties, and the right half correspondingly lists the established equations.

Algorithm 1: Association for the Image Sequence.

1: **Input:** The extracted projection components $\{\vec{\mathbf{p}}_{k,i}\}$, $k = 1, 2$, $i = 1, 2, \ldots, N$, N is the number of images.
2: **Output:** The associated projection components $\{\vec{\mathbf{p}}_{k,i}\}$.
Preliminary association:
3: For j from 2 to N do
3.1 Calculate the CJ operator $\boldsymbol{cJ}_j$ of the jth image;
3.2 If $\boldsymbol{cJ}_j > 0$ swap the values of $\vec{\mathbf{p}}_{1,j}$ and $\vec{\mathbf{p}}_{2,j}$;
Recheck:
4: For j from 1 to N do
4.1 Calculate the RJ operator $\boldsymbol{rJ}_j$ of the jth image;
4.2 If $\boldsymbol{rJ}_j > 0$ swap the values of $\vec{\mathbf{p}}_{1,j}$ and $\vec{\mathbf{p}}_{2,j}$.

C. Establish Over-Determined System of Equations

Compared with the basic imaging projection model (2) used in existing methods, our properties can impose additional constraints on the estimation process. **Properties 1, 2, and 5** constrain the direction of the side vectors, and Property 3 constrains both the directions and lengths. Then, we model the estimation process as an over-determined system $\mathbb{S}_{\mathrm{Comp}}$: $\mathbb{S}_{\mathrm{Comp},L} = \mathbb{S}_{\mathrm{Comp},R}$. The matrices $\mathbb{S}_{\mathrm{Comp},L}$ and $\mathbb{S}_{\mathrm{Comp},R}$ are given in (21) and (22). The dimension of $\mathbb{S}_{\mathrm{Comp},L}$ is $(N^2 + 5N + 1) \times 1$. For clarity, the relationships between the elements of $\mathbb{S}_{\mathrm{Comp}}$ and the proved properties are presented in Fig. 7

$$
\mathbb{S}_{\mathrm{Comp},L} = \begin{bmatrix} \llbracket \mathbf{H}_i\vec{\mathbf{a}} \rrbracket_{i=1,2,\ldots,N} \\ \llbracket \mathbf{H}_i\vec{\mathbf{b}} \rrbracket_{i=1,2,\ldots,N} \\ \vec{\mathbf{a}}^T\vec{\mathbf{b}} \\ \llbracket \left(\sum_{j=1,j\neq i}^{N} \vec{\mathbf{n}}_{z,i,j}^T\right)\vec{\mathbf{a}} \rrbracket_{i=1,2,\ldots,N} \\ \llbracket \left(\sum_{j=1,j\neq i}^{N} \vec{\mathbf{n}}_{z,i,j}^T\right)\vec{\mathbf{b}} \rrbracket_{i=1,2,\ldots,N} \\ \left\llbracket \left\llbracket \begin{matrix} \left(\vec{\mathbf{l}}_{3,i,j}^T\vec{\mathbf{a}}\right)\cdot\left(\vec{\mathbf{n}}_i^T\vec{\mathbf{b}}\right) \\ \cdot\left(\vec{\mathbf{n}}_j^T\left(\vec{\mathbf{a}}\otimes\vec{\mathbf{b}}\right)\right) \end{matrix} \right\rrbracket_{j=i+1,\ldots,N} \right\rrbracket_{i=1,2,\ldots,N-1} \\ \left\llbracket \left\llbracket \begin{matrix} \left(\vec{\mathbf{l}}_{4,i,j}^T\vec{\mathbf{a}}\right)\cdot\left(\vec{\mathbf{n}}_j^T\vec{\mathbf{b}}\right) \\ \cdot\left(\vec{\mathbf{n}}_i^T\left(\vec{\mathbf{a}}\otimes\vec{\mathbf{b}}\right)\right) \end{matrix} \right\rrbracket_{j=i+1,\ldots,N} \right\rrbracket_{i=1,2,\ldots,N-1} \end{bmatrix} \tag{21}
$$

$$
\mathbb{S}_{\mathrm{Comp},R} = \begin{bmatrix} \llbracket \overrightarrow{\mathbf{Ip}}_{1,i} \rrbracket_{i=1,2,\ldots,N} \\ \llbracket \overrightarrow{\mathbf{Ip}}_{2,i} \rrbracket_{i=1,2,\ldots,N} \\ 0 \\ \llbracket \left(\sum_{j=1,j\neq i}^{N} \vec{\mathbf{n}}_{z,i,j}^T\right)\vec{\mathbf{p}}_{1,i} \rrbracket_{i=1,2,\ldots,N} \\ \llbracket \left(\sum_{j=1,j\neq i}^{N} \vec{\mathbf{n}}_{z,i,j}^T\right)\vec{\mathbf{p}}_{2,i} \rrbracket_{i=1,2,\ldots,N} \\ \left\llbracket \left\llbracket \begin{matrix} \left(\vec{\mathbf{l}}_{3,i,j}^T\vec{\mathbf{b}}\right)\cdot\left(\vec{\mathbf{n}}_j^T\vec{\mathbf{a}}\right) \\ \cdot\left(\vec{\mathbf{n}}_i^T\left(\vec{\mathbf{a}}\otimes\vec{\mathbf{b}}\right)\right) \end{matrix} \right\rrbracket_{j=i+1,\ldots,N} \right\rrbracket_{i=1,2,\ldots,N-1} \\ \left\llbracket \left\llbracket \begin{matrix} \left(\vec{\mathbf{l}}_{4,i,j}^T\vec{\mathbf{b}}\right)\cdot\left(\vec{\mathbf{n}}_i^T\vec{\mathbf{a}}\right) \\ \cdot\left(\vec{\mathbf{n}}_j^T\left(\vec{\mathbf{a}}\otimes\vec{\mathbf{b}}\right)\right) \end{matrix} \right\rrbracket_{j=i+1,\ldots,N} \right\rrbracket_{i=1,2,\ldots,N-1} \end{bmatrix}. \tag{22}
$$

In (21) and (22), $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are the side vectors to be solved, $\mathbf{H}_i$ are the imaging projection matrices, $\vec{\mathbf{p}}_{k,i}$ are the associated projection components, $\overrightarrow{\mathbf{Ip}}_{k,i} = \mathbf{H}_i\vec{\mathbf{p}}_{k,i}$. The normal vectors $\vec{\mathbf{n}}_i$ are calculated from $\mathbf{H}_i$, and the intersection lines $\vec{\mathbf{l}}_{3,i,j}, \vec{\mathbf{l}}_{4,i,j}$ are calculated from $\vec{\mathbf{p}}_{k,i}$ and $\vec{\mathbf{p}}_{k,j}$. The details of calculation are given in Algorithm 2. $\llbracket f(i)\rrbracket_i$ represents arranging $f(i)$ vertically in ascending order of i to form a matrix. An example is given as

$$
\left[\mathbf{H}_i\vec{\mathbf{a}}\right]_{i=1,2,\ldots,N} = \begin{bmatrix} \mathbf{H}_1\vec{\mathbf{a}} \\ \ldots \\ \mathbf{H}_i\vec{\mathbf{a}} \\ \ldots \\ \mathbf{H}_N\vec{\mathbf{a}} \end{bmatrix}_{2N\times 1}. \tag{23}
$$

D. Solve Overdetermined System of Equations

As shown in (21) and (22), the side vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are inherently intertwined in the established system $\mathbb{S}_{\mathrm{Comp}}$. It is challenging to directly derive its optimal solution. In our work, a coarse estimation is implemented in advance to assist in decoupling this system into linear subsystems, and these subsystems are solved with the alternating iterative strategy.

1) *Coarse Estimation With a WS:* Based on the basic imaging projection model (2), two independent and linear systems of equations about side vectors can be established. However, the constraints of multiple geometric properties are absent in these systems, and the estimation accuracy can hardly be ensured. To impose these constraints without introducing coupled side vectors into equations, we adopt a WS. Then, the weighted linear systems are solved by the Least Square (LS) algorithm.

a) *Determine weights:* The projection extraction accuracies vary across images. Some projection components can negatively impact the estimation process. To alleviate this issue, we define a measurement operator to assess the suitability of the projection components for effective estimation.

Properties 2, 3, and **5** establish constraints on the geometric relationships among the projection components. When these projection components are error-free, they fully conform to these geometric relationships, meaning the two sides of the equations in the properties are exactly equal. Under these conditions, the estimation results are also free of errors. Conversely, projection extraction errors can lead to discrepancies between the projection components and theoretical relationships, resulting in estimation errors. When the projection components are substituted into the equations in the properties (‘PE’s established in Section III), a smaller residual of the equation indicates a higher fit degree between the projection components and theoretical relationships. The projection components with smaller residuals are considered more suitable for estimation. Therefore, the measurement operator is defined as the reciprocal of the equation residuals.

By substituting projection components $\vec{\mathbf{p}}_{k,i}$ and $\vec{\mathbf{p}}_{k,j}$ into the equations, the residuals concerning Properties 2, 3, and 5 are, respectively, calculated by

$$
\chi_{i,j,1} = \left|PE_{2,1,i,j}^{L}\right|. \tag{24}
$$

$$
\chi_{i,j,2} = \sum_{h=1}^{2}\mathcal{D}\left(PE_{3,h,i,j}^{L}, PE_{3,h,i,j}^{R}\right). \tag{25}
$$

$$
\chi_{i,j,3} = \sum_{h=1}^{2}\mathcal{D}\left(PE_{5,h,i,j}^{L}, PE_{5,h,i,j}^{R}\right). \tag{26}
$$

In ‘PE’s, the vectors $\vec{\mathbf{n}}_i$, $\vec{\mathbf{n}}_j$, $\vec{\mathbf{n}}_{z,i,j}$, $\vec{\mathbf{l}}_{1/2/3/4,i,j}$, and $\vec{\mathbf{z}}$ are calculated from the imaging projection matrices and associated projection components. The details of calculation are given in Algorithm 2.

The measurement operator $\mathcal{M}_i$ for the ith image is calculated by

$$
\begin{aligned} \mathcal{M}_i &= \frac{1}{\chi_i + 10^{-5}} \\ \chi_i &= \sum_{j=1,j\neq i}^{N} \chi_{i,j,1} + \beta_1\cdot\chi_{i,j,2} + \beta_2\cdot\chi_{i,j,3} \end{aligned} \tag{27}
$$

where β_1 and β_2 are manually set constants.

By normalizing $\mathcal{M}_i$, the weight coefficient ζ_i for the ith image is defined as

$$\zeta_i = \frac{\mathcal{M}_i}{\sum_{i=1}^{N} \mathcal{M}_i}. \tag{28}$$

b) *Solve the weighed system:* The weighted linear systems for the side vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are represented as

$$\left[[\zeta_i \cdot \mathbf{H}_i]_{i=1,2,\ldots,N}\right] \vec{\mathbf{a}} = \left[\left[\!\left[\zeta_i \cdot \overrightarrow{\mathbf{Ip}}_{1,i}\right]\!\right]_{i=1,2,\ldots,N}\right] \tag{29}$$

$$\left[[\zeta_i \cdot \mathbf{H}_i]_{i=1,2,\ldots,N}\right] \vec{\mathbf{b}} = \left[\left[\!\left[\zeta_i \cdot \overrightarrow{\mathbf{Ip}}_{2,i}\right]\!\right]_{i=1,2,\ldots,N}\right]. \tag{30}$$

These systems are solved by the LS algorithm

$$\vec{\mathbf{e}}_{k,0} = \left[[\zeta_i \cdot \mathbf{H}_i]_{i=1,2,\ldots,N}\right]_{2N\times 3} \backslash \left[\left[\!\left[\zeta_i \cdot \overrightarrow{\mathbf{Ip}}_{k,i}\right]\!\right]_{i=1,2,\ldots,N}\right]_{2N\times 1} \tag{31}$$

where $\vec{\mathbf{e}}_{1,0}$ and $\vec{\mathbf{e}}_{2,0}$ are, respectively, the initial solution for the side vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$. The left division for matrices is calculated by $\mathbf{A}\backslash\mathbf{B} = \mathbf{A}^{+}\mathbf{B}$, and $\mathbf{A}^{+}$ is the pseudoinverse of $\mathbf{A}$ obtained by the singular value decomposition. Notably, to ensure that the system of equations has a unique LS solution, the number of rows in the matrix $[[\![\zeta_i \cdot \mathbf{H}_i]\!]_{i=1,2,\ldots,N}]$ must exceed 3 ($2N > 3$) and this matrix should be of full column rank. In practical terms, this means that at least two images are required, and the normal vectors of their imaging planes must differ. This requirement aligns with *Assumption 2* in Section III.

The WS retains the constraints of multiple geometric properties, which ensures the accuracy of our coarse estimation.

2) *Decouple the System Into Subsystems:* To decouple the complex system, we introduce an auxiliary vector $\vec{\mathbf{z}} = \vec{\mathbf{a}} \otimes \vec{\mathbf{b}}/(\|\vec{\mathbf{a}}\|_2 \cdot \|\vec{\mathbf{b}}\|_2)$, and its relationships with the side vectors are constrained by (32). The initial solution $\vec{\mathbf{z}}_0$ for $\vec{\mathbf{z}}$ is given as

$$\begin{aligned} \vec{\mathbf{z}}^T \vec{\mathbf{a}} &= 0 \\ \vec{\mathbf{z}}^T \vec{\mathbf{b}} &= 0 \\ \left(\vec{\mathbf{a}} \otimes \vec{\mathbf{b}} / \left(\|\vec{\mathbf{a}}\|_2 \cdot \left\|\vec{\mathbf{b}}\right\|_2\right)\right)^T \vec{\mathbf{z}} &= 1 \end{aligned} \tag{32}$$

$$\vec{\mathbf{z}}_0 = \vec{\mathbf{e}}_{1,0} \otimes \vec{\mathbf{e}}_{2,0} / \left(\left\|\vec{\mathbf{e}}_{1,0}\right\|_2 \cdot \left\|\vec{\mathbf{e}}_{2,0}\right\|_2\right). \tag{33}$$

When $\vec{\mathbf{b}}$ and $\vec{\mathbf{z}}$ are fixed as their initial solutions $\vec{\mathbf{e}}_{2,0}$ and $\vec{\mathbf{z}}_0$, the system $\mathbb{S}_{\text{Comp}}$ is equivalent to a linear over-determined system $\mathbb{S}_A$ for the side $\vec{\mathbf{a}}$. The subsystem $\mathbb{S}_A$ can be represented as $\mathbf{H}_A \vec{\mathbf{a}} = \vec{v}_A$, where the matrix $\mathbf{H}_A$ and vector $\vec{v}_A$ are given as

$$\mathbf{H}_A = \begin{bmatrix} [\![\mathbf{H}_i]\!]_{i=1,2,\ldots,N} \\ \vec{\mathbf{e}}_{2,0}^T \\ \left[\!\left[\sum_{j=1,j\neq i}^{N} \vec{\mathbf{n}}_{z,i,j}^T\right]\!\right]_{i=1,2,\ldots,N} \\ \left[\!\left[\left[\!\left[\begin{matrix} \left(\vec{\mathbf{n}}_i^T \vec{\mathbf{e}}_{2,0}\right) \cdot \left(\vec{\mathbf{n}}_j^T \vec{\mathbf{z}}_0\right) \vec{\mathbf{l}}_{3,i,j}^T \\ - \left(\vec{\mathbf{l}}_{3,i,j}^T \vec{\mathbf{e}}_{2,0}\right) \cdot \left(\vec{\mathbf{n}}_i^T \vec{\mathbf{z}}_0\right) \vec{\mathbf{n}}_j^T \end{matrix}\right]\!\right]_{j=i+1,\ldots,N}\right]\!\right]_{i=1,2,\ldots,N-1} \\ \left[\!\left[\left[\!\left[\begin{matrix} \left(\vec{\mathbf{n}}_j^T \vec{\mathbf{e}}_{2,0}\right) \cdot \left(\vec{\mathbf{n}}_i^T \vec{\mathbf{z}}_0\right) \vec{\mathbf{l}}_{4,i,j}^T \\ - \left(\vec{\mathbf{l}}_{4,i,j}^T \vec{\mathbf{e}}_{2,0}\right) \cdot \left(\vec{\mathbf{n}}_j^T \vec{\mathbf{z}}_0\right) \vec{\mathbf{n}}_i^T \end{matrix}\right]\!\right]_{j=i+1,\ldots,N}\right]\!\right]_{i=1,2,\ldots,N-1} \\ \vec{\mathbf{z}}_0^T \end{bmatrix} \tag{34}$$

$$\vec{v}_A = \begin{bmatrix} \left[\!\left[\overrightarrow{\mathbf{Ip}}_{1,i}\right]\!\right]_{i=1,2,\ldots,N} \\ 0 \\ \left[\!\left[\left(\sum_{j\neq i} \vec{\mathbf{n}}_{z,i,j}^T\right) \vec{\mathbf{p}}_{1,i}\right]\!\right]_{i=1,2,\ldots,N} \\ \vec{\mathbf{0}}_{(N^2-N+1)\times 1} \end{bmatrix} \tag{35}$$

where $\vec{\mathbf{0}}_{(N^2-N)\times 1}$ is a zero-value column vector. The calculation process of $\vec{\mathbf{n}}_i$, $\vec{\mathbf{n}}_j$, $\vec{\mathbf{n}}_{z,i,j}$, $\vec{\mathbf{l}}_{3,i,j}$, and $\vec{\mathbf{l}}_{4,i,j}$ is given in Algorithm 2. The dimension of $\mathbf{H}_A$ is $(N^2 + 2N + 2) \times 3$.

Similarly, the linear over-determined system $\mathbb{S}_B$ for the side $\vec{\mathbf{b}}$ is $\mathbf{H}_B \vec{\mathbf{b}} = \vec{v}_B$, where the matrix $\mathbf{H}_B$ and vector $\vec{v}_B$ are given as

$$\mathbf{H}_B = \begin{bmatrix} [\![\mathbf{H}_i]\!]_{i=1,2,\ldots,N} \\ \vec{\mathbf{e}}_{1,0}^T \\ \left[\!\left[\sum_{j=1,j\neq i}^{N} \vec{\mathbf{n}}_{z,i,j}^T\right]\!\right]_{i=1,2,\ldots,N} \\ \left[\!\left[\left[\!\left[\begin{matrix} \left(\vec{\mathbf{l}}_{3,i,j}^T \vec{\mathbf{e}}_{1,0}\right) \cdot \left(\vec{\mathbf{n}}_j^T \vec{\mathbf{z}}_0\right) \vec{\mathbf{n}}_i^T \\ - \left(\vec{\mathbf{n}}_j^T \vec{\mathbf{e}}_{1,0}\right) \cdot \left(\vec{\mathbf{n}}_i^T \vec{\mathbf{z}}_0\right) \vec{\mathbf{l}}_{3,i,j}^T \end{matrix}\right]\!\right]_{j=i+1,\ldots,N}\right]\!\right]_{i=1,2,\ldots,N-1} \\ \left[\!\left[\left[\!\left[\begin{matrix} \left(\vec{\mathbf{l}}_{4,i,j}^T \vec{\mathbf{e}}_{1,0}\right) \cdot \left(\vec{\mathbf{n}}_i^T \vec{\mathbf{z}}_0\right) \vec{\mathbf{n}}_j^T \\ - \left(\vec{\mathbf{n}}_i^T \vec{\mathbf{e}}_{1,0}\right) \cdot \left(\vec{\mathbf{n}}_j^T \vec{\mathbf{z}}_0\right) \vec{\mathbf{l}}_{4,i,j}^T \end{matrix}\right]\!\right]_{j=i+1,\ldots,N}\right]\!\right]_{i=1,2,\ldots,N-1} \\ \vec{\mathbf{z}}_0^T \end{bmatrix} \tag{36}$$

$$\vec{v}_B = \begin{bmatrix} \left[\!\left[\overrightarrow{\mathbf{Ip}}_{2,i}\right]\!\right]_{i=1,2,\ldots,N} \\ 0 \\ \left[\!\left[\left(\sum_{j\neq i} \vec{\mathbf{n}}_{z,i,j}^T\right) \vec{\mathbf{p}}_{2,i}\right]\!\right]_{i=1,2,\ldots,N} \\ \vec{\mathbf{0}}_{(N^2-N+1)\times 1} \end{bmatrix}. \tag{37}$$

Notably, the auxiliary vector $\vec{\mathbf{z}}$ is also updated during the iterations. When $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are fixed as $\vec{\mathbf{e}}_{1,0}$ and $\vec{\mathbf{e}}_{2,0}$, the system $\mathbb{S}_{\text{Comp}}$ is equivalent to a linear system $\mathbb{S}_Z$ for $\vec{\mathbf{z}}$, which is represented as $\mathbf{H}_Z \vec{\mathbf{z}} = \vec{v}_Z$. The dimension of $\mathbf{H}_Z$

is $(N^2-N+1)\times 3$

$$\mathbf{H}_Z = \begin{bmatrix} \left[\!\left[\left[\!\left[\begin{matrix} \left(\vec{\mathbf{l}}_{3,i,j}^T \vec{\mathbf{e}}_{1,0}\right)\cdot\left(\vec{\mathbf{n}}_i^T \vec{\mathbf{e}}_{2,0}\right)\vec{\mathbf{n}}_j^T \\ -\left(\vec{\mathbf{n}}_j^T \vec{\mathbf{e}}_{1,0}\right)\cdot\left(\vec{\mathbf{l}}_{3,i,j}^T \vec{\mathbf{e}}_{2,0}\right)\vec{\mathbf{n}}_i^T \end{matrix} \right]\!\right]_{j=i+1,\ldots,N} \right]\!\right]_{i=1,2,\ldots,N-1} \\ \left[\!\left[\left[\!\left[\begin{matrix} \left(\vec{\mathbf{l}}_{4,i,j}^T \vec{\mathbf{e}}_{1,0}\right)\cdot\left(\vec{\mathbf{n}}_j^T \vec{\mathbf{e}}_{2,0}\right)\vec{\mathbf{n}}_i^T \\ -\left(\vec{\mathbf{n}}_i^T \vec{\mathbf{e}}_{1,0}\right)\cdot\left(\vec{\mathbf{l}}_{4,i,j}^T \vec{\mathbf{e}}_{2,0}\right)\vec{\mathbf{n}}_j^T \end{matrix} \right]\!\right]_{j=i+1,\ldots,N} \right]\!\right]_{i=1,2,\ldots,N-1} \\ \left(\vec{\mathbf{e}}_{1,0}\otimes\vec{\mathbf{e}}_{2,0}/\left(\left\|\vec{\mathbf{e}}_{1,0}\right\|_2\cdot\left\|\vec{\mathbf{e}}_{2,0}\right\|_2\right)\right)^T \end{bmatrix} \tag{38}$$

$$\vec{v}_Z = \begin{bmatrix} \vec{\mathbf{0}}_{(N^2-N)\times 1} \\ 1 \end{bmatrix}. \tag{39}$$

3) *Solve the Subsystems by AIRLS Algorithm:* The Iterative Reweighted Least Square (IRLS) algorithm is a common approach for solving linear over-determined systems of equations, which exhibits strong robustness against outliers in data [28].

Given that some extracted projection components may have significantly lower accuracy, we also introduce the iterative reweighting strategy to mitigate the impact of these abnormal projection components on the estimation results. Then, the objective function for solving the original system $\mathbb{S}_{\text{Comp}}$ is established as

$$F = \min_{\vec{\mathbf{a}},\vec{\mathbf{b}},\vec{\omega}} \left(\boldsymbol{\Psi}\overrightarrow{\mathbf{rs}}\right)^T\left(\boldsymbol{\Psi}\overrightarrow{\mathbf{rs}}\right)$$
$$\text{s.t.}\|\vec{\omega}\|_1 = \sigma,\ \overrightarrow{\mathbf{rs}} = \mathbb{S}_{\text{Comp},L} - \mathbb{S}_{\text{Comp},R} \tag{40}$$

where $\overrightarrow{\mathbf{rs}}$ is the residual vector of equations, $\vec{\omega} = [\![\omega_m]\!]_{m=1,2,\ldots,M}$ is the weight vector, M is the number of equations, ω_m should satisfy $\omega_m > 0$, $\boldsymbol{\Psi} = \text{diag}(\vec{\omega})$, and $\text{diag}(\cdot)$ is a function to generate a diagonal matrix.

Due to the intertwined side vectors in the complex system $\mathbb{S}_{\text{Comp}}$, the objective function (40) is nonconvex. It is time-consuming to solve the system by conventional optimization algorithms, such as the Particle Swarm Optimization (PSO) algorithm.

Combining with the decoupled linear subsystems, we design the AIRLS algorithm to avoid the time-consuming optimizations. Its details are listed further.

a) *Derive update formulas in iterations:* In the kth ($k \in \mathrm{N}^+$) iteration, the update formulas are derived as follows.

When updating $\vec{\mathbf{a}}$, the vectors $\vec{\mathbf{b}}$, $\vec{\mathbf{z}}$, and $\vec{\omega}$ are fixed. The system $\mathbb{S}_{\text{Comp}}$ is equivalent to $\mathbb{S}_A$. Then, its optimum is derived by the LS algorithm

$$\vec{\mathbf{a}}^{(k)} = (\boldsymbol{\Psi}\mathbf{H}_A)\setminus\left(\boldsymbol{\Psi}\vec{v}_A\right). \tag{41}$$

Similarly, we can obtain the update formula for $\vec{\mathbf{b}}$ and $\vec{\mathbf{z}}$

$$\vec{\mathbf{b}}^{(k)} = (\boldsymbol{\Psi}\mathbf{H}_B)\setminus\left(\boldsymbol{\Psi}\vec{v}_B\right) \tag{42}$$
$$\vec{\mathbf{z}}^{(k)} = (\boldsymbol{\Psi}_Z\mathbf{H}_Z)\setminus\left(\boldsymbol{\Psi}_Z\vec{v}_Z\right). \tag{43}$$

For $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$, we utilize the same weight vectors $\vec{\omega} \in \mathbb{R}^{(N^2+2N+2)\times 1}$. The weight vector $\vec{\omega}_Z$ for $\vec{\mathbf{z}}$ is a part of the

Algorithm 2: Auto-Correction Based on AIRLS.

1: **Input:** $\mathbf{H}_A$, $\vec{v}_A$, $\mathbf{H}_B$, $\vec{v}_B$, $\mathbf{H}_Z$, $\vec{v}_Z$, $\boldsymbol{\Psi}$, $\vec{\mathbf{e}}_{1,0}$, and $\vec{\mathbf{e}}_{2,0}$.
2: **Output:** The side vectors $\vec{\mathbf{a}}, \vec{\mathbf{b}} \in \mathbb{R}^{3\times 1}$.
3: **Initialize:** $\Delta e = 1$, $\vec{\mathbf{a}}^{(0)} = \vec{\mathbf{e}}_{1,0}$, $\vec{\mathbf{b}}^{(0)} = \vec{\mathbf{e}}_{2,0}$, $k = 0$.
4: While $\Delta e \geq 10^{-4}$ do:
4.1 $k = k + 1$.
4.2 Calculate $\vec{\mathbf{a}}^{(k)}$ by (41).
4.3 Replace$\vec{\mathbf{e}}_{1,0}$ in $\mathbf{H}_B$ and $\mathbf{H}_Z$ with $\vec{\mathbf{a}}^{(k)}$.
4.4 Calculate $\vec{\mathbf{b}}^{(k)}$ by (42).
4.5 Replace$\vec{\mathbf{e}}_{2,0}$ in $\mathbf{H}_A$ and $\mathbf{H}_Z$ with $\vec{\mathbf{b}}^{(k)}$.
4.6 Calculate $\vec{\mathbf{z}}^{(k)}$ by (43).
4.7 Replace$\vec{\mathbf{z}}_0$ in $\mathbf{H}_A$ and $\mathbf{H}_B$ with $\vec{\mathbf{z}}^{(k)}$.
4.8 Update the weights by (46) and (47).
4.9 Calculate the change Δe by (50).

weight vector $\vec{\omega}${{{\vec{\bf{\omega }}}}_Z}

$$\vec{\omega}_Z = [\omega_m]_{\substack{m\text{from}3N+2\\ \text{to}N^2+2N+2}}. \tag{44}$$

When updating the weights $\vec{\omega}$, the residual vector $\overrightarrow{\mathbf{rs}}$ is fixed. The update formula for weights in our algorithm is consistent with that in the IRLS algorithm

$$\vec{\omega}^{(k)} = \arg\min_{\vec{\omega}} \left(\boldsymbol{\Psi}\overrightarrow{\mathbf{rs}}\right)^T\left(\boldsymbol{\Psi}\overrightarrow{\mathbf{rs}}\right) - \lambda\left(\|\vec{\omega}\|_1 - \sigma\right)$$
$$= \arg\min_{\vec{\omega}} \sum_{m=1}^{M} rs_m^2\left(\omega_m - \frac{\lambda}{2rs_m^2}\right)^2 + \sigma' \tag{45}$$

where rs_m is the mth element of $\overrightarrow{\mathbf{rs}}$, λ and σ are manually set constants. Therefore, the $\vec{\omega}$ can be updated by the reciprocal of the residual

$$\omega_m^{(k)} = \frac{1}{rs_m^2 + 10^{-5}}. \tag{46}$$

In our algorithm, rs_m^2 is calculated as

$$\overrightarrow{\mathbf{rs}}_A = \mathbf{H}_A\vec{\mathbf{a}}^{(k)} - \vec{v}_A$$
$$\overrightarrow{\mathbf{rs}}_B = \mathbf{H}_B\vec{\mathbf{b}}^{(k)} - \vec{v}_B$$
$$rs_m^2 = rs_{\text{A},m}^2 + rs_{\text{B},m}^2. \tag{47}$$

b) *Analyze the convergence:* Since $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$, and $\vec{\mathbf{z}}$ are updated by the LS solution, we can obtain

$$F\left(\vec{\mathbf{a}}^{(k-1)}, \vec{\mathbf{b}}^{(k-1)}, \vec{\mathbf{z}}^{(k-1)}, \vec{\omega}^{(k-1)}\right) >$$
$$F\left(\vec{\mathbf{a}}^{(k)}, \vec{\mathbf{b}}^{(k-1)}, \vec{\mathbf{z}}^{(k-1)}, \vec{\omega}^{(k-1)}\right) >$$
$$F\left(\vec{\mathbf{a}}^{(k)}, \vec{\mathbf{b}}^{(k)}, \vec{\mathbf{z}}^{(k-1)}, \vec{\omega}^{(k-1)}\right) > F\left(\vec{\mathbf{a}}^{(k)}, \vec{\mathbf{b}}^{(k)}, \vec{\mathbf{z}}^{(k)}, \vec{\omega}^{(k-1)}\right). \tag{48}$$

According to (45), the objective function for $\vec{\omega}$ is a convex function when the residual is fixed. Therefore, $F(\vec{\mathbf{a}}^{(k)}, \vec{\mathbf{b}}^{(k)}, \vec{\mathbf{z}}^{(k)}, \vec{\omega}^{(k-1)}) > F(\vec{\mathbf{a}}^{(k)}, \vec{\mathbf{b}}^{(k)}, \vec{\mathbf{z}}^{(k)}, \vec{\omega}^{(k)})$. Then, the convergence of the algorithm is proved.

c) *Initialization:* The subsystems $\mathbb{S}_A$, $\mathbb{S}_B$, and $\mathbb{S}_Z$ are initialized in Section IV-D1. The weight matrix $\boldsymbol{\Psi}$ is

TABLE III
Radar and Imaging Parameters

Radar Location	Hangzhou (120.2E,30.3N,0m)	
Radar Parameters	Center Frequency	16.67 GHz
	PRF	16 Hz
	Bandwidth	4 GHz
Imaging Parameters	CPI	512 Pulses
	Imaging Interval	10 s

initialized as

$$\boldsymbol{\Psi} = \mathrm{diag}\left(\begin{bmatrix} [\![[\![\begin{matrix}\zeta_i \\ \zeta_i\end{matrix}]\!]]\!]_{i=1,2,\ldots,N} \\ 1 \\ [\zeta_i]_{i=1,2,\ldots,N} \\ [\![[\![\zeta_i\zeta_j]\!]_{j=i+1,\ldots,N}]\!]_{i=1,2,\ldots,N-1} \\ [\![[\zeta_i\zeta_j]_{j=i+1,\ldots,N}]\!]_{i=1,2,\ldots,N-1} \\ 1 \end{bmatrix}\right). \tag{49}$$

where ζ_i the weight coefficients are obtained by (28).

d) *Termination condition:* The change Δe of side vectors in the kth iteration is defined as

$$\Delta e = \frac{\left\|\vec{\mathbf{a}}^{(k)} - \vec{\mathbf{a}}^{(k-1)}\right\|_2}{\left\|\vec{\mathbf{a}}^{(k-1)}\right\|_2} + \frac{\left\|\vec{\mathbf{b}}^{(k)} - \vec{\mathbf{b}}^{(k-1)}\right\|}{\left\|\vec{\mathbf{b}}^{(k-1)}\right\|_2}. \tag{50}$$

The iteration terminates when $\Delta e < 10^{-4}$.

The AIRLS algorithm is summarized as ***Algorithm 2***. The absolute attitude and 3D size of the solar panel are determined by the algorithm output, i.e., the final delicate solution for the complex system $\mathbb{S}_{\text{Comp}}$. Our ALRLS algorithm avoids the time-consuming optimizations.

V. Experimental Analysis

In this section, some experiments are carried out to illustrate the performance of the proposed method. First, estimation accuracies under different observation conditions and SNRs are analyzed by experiments on radar images. The side-based method [4], the PCA-based method [8], and the QSVP-based method [10] are selected as comparative methods. Then, the applicability of our method to optical images is verified. In this experiment, the QSVP-based method [10] is not applicable.

A. Experiments on Radar Images

The experimental conditions are established based on realistic space observation scenarios. We utilize a Ku-band radar, whose parameters are listed in Table III, to generate echoes. The Aura satellite, whose 3D model is shown in Fig. 9(a), is a three-axis stabilized target. In the RTN system, its attitude remains unchanged. The attitude and size estimation error is defined as

$$E_{\text{attitude}} = \frac{1}{2}\sum_{k=1}^{2} \angle\left(\vec{\mathbf{e}}_{t,k}, \vec{\mathbf{e}}_{e,k}\right) \tag{51}$$

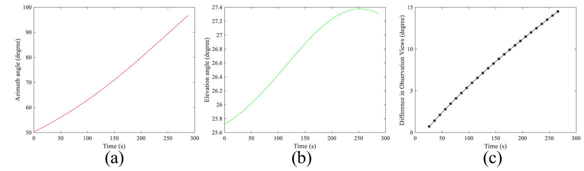

Fig. 8. Curves of observation angles. (a) Azimuth angles. (b) Elevation angles. (c) DOV among the image sequence.

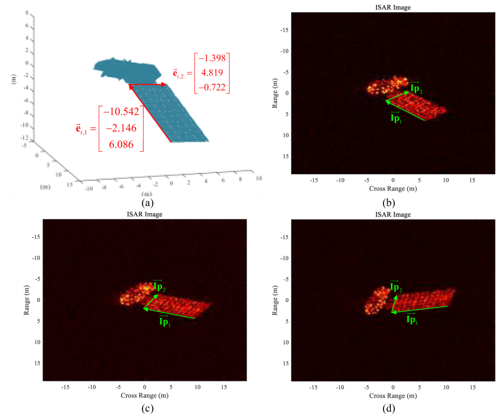

Fig. 9. 3D model and simulated ISAR images of the Aura satellite. The red vectors are 3D vectors of solar panel sides. The green vectors are projections extracted from images.

$$E_{\text{size}} = \frac{1}{2}\sum_{k=1}^{2} \left| \left\|\vec{\mathbf{e}}_{t,k}\right\|_2 - \left\|\vec{\mathbf{e}}_{e,k}\right\|_2 \right| \tag{52}$$

where $\angle(\cdot)$ represents the angle between vectors, $\vec{\mathbf{e}}_{t,k}$ ($k = 1, 2$) are the two actual side vectors of its solar panel, and $\vec{\mathbf{e}}_{e,k}$ ($k = 1, 2$) are the estimated side vectors. In calculation, $\vec{\mathbf{e}}_{t,1}$ and $\vec{\mathbf{e}}_{e,1}$ are vectors corresponding to the longer side, $\vec{\mathbf{e}}_{t,2}$ and $\vec{\mathbf{e}}_{e,2}$ are corresponding to the shorter side. The units of attitude and size estimation errors are, respectively, "degrees" and "meters.". To evaluate the size estimation accuracy, most existing works utilize the absolute error instead of relative error to avoid the impact of scale variations [4], [5], [6], [7], [8], [9], [10].

With the actual radar location and satellite orbital elements, we can obtain the observation angles in the ENU system. Then, the observation angles in the RTN system, which are shown in Fig. 8(a) and (b), can be further calculated via coordinate transformation [22]. The difference in observation views (DOVs) among the image sequence is defined as (53), and its variation curves are given in Fig. 8(c)

$$DOV = \max_{i,j} \angle\left(\vec{\mathbf{n}}_i, \vec{\mathbf{n}}_j\right) \tag{53}$$

where $\vec{\mathbf{n}}_i$ is the normal vector of the ith imaging plane, which is calculated from the projection matrix $\mathbf{H}_i$ by (79).

With the target 3D model and observation angles, we generate simulated radar by the Physical Optics algorithm [30], and well-focused and scaled ISAR images are obtained by the modified Range-Doppler imaging algorithms [31], [32], [33]. Some simulated ISAR images are shown in Fig. 9(b)–(d). The parallelogram features used in our

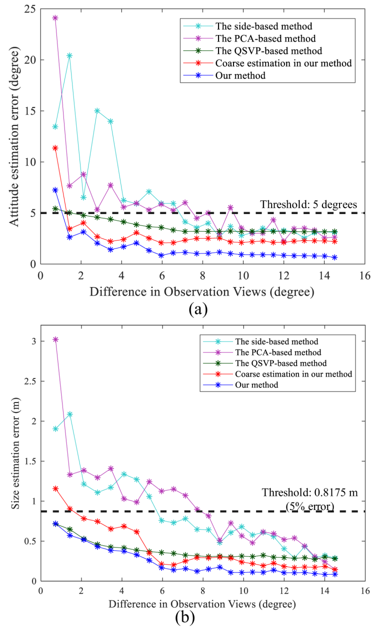

Fig. 10. Estimation errors with respect to DOV. (a) Size estimation errors. (b) Attitude estimation errors. The dotted lines show estimation errors of the methods. The dashed lines give the thresholds for reliable estimations.

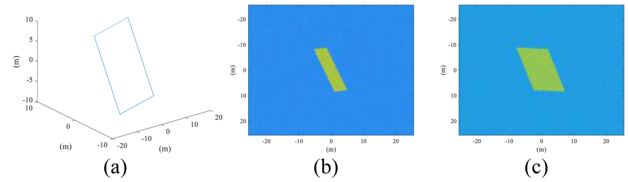

Fig. 11. Diagram of the generated planar rectangle and images. (a) Planar rectangle. (b) First image. (c) Second image ($\theta_D = 30^\circ$).

method and comparative methods are extracted by the image processing operations described in [4]. The extracted projections are labeled green in Fig. 9.

1) *Analyze Estimation Accuracy Under Different Observation Conditions:* In this experiment, the estimation is carried out by making use of all the images currently observed. The SNR of the observed echoes is 0 dB. As shown in Fig. 8(c), the DOV increases along with observation time. Then, the estimation accuracies under different DOVs can be analyzed, and their variation curves are given in Fig. 10. According to [4], [5], [6], [7], [8], [9], [10], when attitude and size estimation errors are, respectively, below 5° and 5%, the estimation results can be considered to be reliable.

Our method and the QSVP-based method can achieve reliable estimation results when the DOV is larger than 2° (70 s), while the other existing methods require the DOV to reach 9° (170 s). Notably, when the DOV is between 2° and 6°, compared with the coarse estimation in our method, the QSVP-based method achieves lower size estimation errors and higher attitude estimation errors, which indicates their similar performance. Instead of exploring the imaging geometric properties, the QSVP-based method utilizes the additional information provided by phases of complex-value echoes to promise its estimation accuracy [13]. In this experiment, it is difficult to evaluate which strategy is more effective. However, the estimation accuracy of the QSVP phases cannot be promised in practical applications [13]. Therefore, we will discuss their noise-robustness in the next experiment. Besides, the estimation errors of our final delicate solutions are 1.314° and 0.159 m lower than those of our coarse solutions. This indicates that the strategy of establishing an over-determined system imposes stronger constraints, compared with the selective WS.

2) *Analyze Estimation Accuracy Under Different SNRs:* First, we analyze the estimation errors obtained with limited observations (DOV =2.118°, 62 s, 4 images). As shown in Table IV, the QSVP-based method achieves lower estimation errors than the other two existing methods when SNR is higher than -5 dB. However, its estimation accuracy is the worst at -10 dB, which indicates its poor noise-robustness. Our methods not only ensure the estimation accuracy with limited observations, but also have better noise-robustness. From -10 to 5 dB, the estimation errors of our methods are constantly lower than those of the existing methods. Since the side-based method and the PCA-based method do not utilize additional information, such as geometric properties or spatial-variant phases, their estimation errors are approximately twice those of our method.

Then, we analyze the estimation errors obtained with abundant observations (DOV =8.822°, 172 s, 15 images). As shown in Table V, abundant observations can improve estimation accuracies of all the methods, which also diminish the importance of additional information. As a result, the QSVP-based method no longer has significant advantages compared with the other methods. Notably, since the PCA features are more noise-robust than the side features, our coarse estimation achieves worse estimation accuracies than the PCA-based method at -8 and -10dB. By imposing stronger constraints on the estimation, our final delicate solution achieves the lowest estimation errors under all SNRs.

3) *Analyze Accuracy of Projection Association:* The association of side projections is carried out in both our method and the side-based method [4]. Differently, our association is based on the multiple geometric properties, and the association in [4] is based on the nearest neighbor principle. To compare the effectiveness of these two strategies, we conduct the following Monte Carlo experiments. In each experiment, a planar rectangle is randomly generated in a 3D coordinate system. Then, we generate two images by projecting the rectangle onto two imaging planes, whose normal vectors are $[1, 0, 0]^T$ and $[\cos\theta_D, sin\theta_D, 0]^T$, θ_D is the DOV between two images. The SNR of images is 0 dB.

For example, a generated planar rectangle and the corresponding images are given in Fig. 11.

For each θ_D, the average association accuracy is defined as

$$\mathrm{rate}_{As} = \frac{N_{A-t}}{N_{M-C}} \tag{54}$$

where $N_{M-C} = 100$ is the number of experiments, and N_{A-t} is the number of experiments in which side projections from

TABLE IV
Estimation Errors Obtained With Limited Radar Observations Under Different SNRs (DOV =2.118°)

	Attitude estimation errors (degree)					Size estimation errors (m)				
SNR (dB)	-10	-8	-5	0	5	-10	-8	-5	0	5
Side-based method	20.855	14.522	10.201	6.526	4.066	10.189	6.082	2.277	1.212	0.726
PCA-based method	18.313	15.252	11.423	8.791	7.055	9.518	5.564	2.008	1.385	1.195
QPSV-based method	35.723	18.316	10.265	4.767	2.668	10.988	4.864	1.855	0.528	0.385
Coarse estimation in our method	14.218	9.123	6.567	4.033	2.269	7.867	3.003	1.209	0.782	0.364
Our method	**11.352**	**8.483**	**5.403**	**3.144**	**1.785**	**5.225**	**2.085**	**0.808**	**0.517**	**0.235**

The bold values emphasize that our method achieves lowest estimation errors in experiments under various SNRs.

TABLE V
Estimation Errors Obtained With Abundant Radar Observations Under Different SNRs (DOV =8.822°)

	Attitude estimation errors (degree)					Size estimation errors (m)				
SNR (dB)	-10	-8	-5	0	5	-10	-8	-5	0	5
Side-based method	15.271	10.797	7.529	2.672	1.955	8.192	3.563	1.317	0.482	0.205
PCA-based method	9.558	7.192	6.260	2.967	2.499	6.323	2.385	1.029	0.510	0.419
QPSV-based method	30.723	15.378	8.177	3.206	1.968	8.258	3.955	1.216	0.312	0.313
Coarse estimation in our method	13.256	8.144	4.644	2.546	1.038	6.965	3.291	0.982	0.308	0.194
Our method	**8.988**	**5.662**	**2.558**	**1.168**	**0.678**	**5.587**	**2.243**	**0.603**	**0.175**	**0.088**

The bold values emphasize that our method achieves lowest estimation errors in experiments under various SNRs.

TABLE VI
Errors Obtained by Coarse Estimation With and Without WS

Observation condition		DOV= 2.118°					DOV= 8.822°				
SNR (dB)		-10	-8	-5	0	5	-10	-8	-5	0	5
Attitude error (degree)	Without WS	19.074	11.561	10.811	5.817	3.058	15.123	10.307	6.287	2.615	1.322
	With WS	14.218	9.123	6.567	4.033	2.269	13.256	8.144	4.644	2.546	1.038
Size error (m)	Without WS	8.207	5.927	2.078	1.003	0.523	7.874	3.488	1.299	0.457	0.202
	With WS	7.867	3.003	1.209	0.782	0.364	6.965	3.291	0.982	0.308	0.194

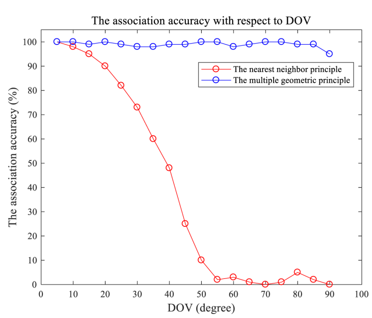

Fig. 12. Association accuracy with respect to DOV.

TABLE VII
Relationship Between NIC and Number of Images

Number of images	4	8	12	16
Average NIC	6.83	6.54	6.85	6.62
Maximum NIC	15	12	14	14
Minimum NIC	4	4	4	4

two images are associated correctly. Here is the variation curve of association accuracy with respect to DOV.

As shown in Fig. 12, the nearest neighbor association may fail when the DOV is large, especially when it is greater than 20 degrees. In contrast, our association algorithm achieves a high association accuracy under all DOVs. Notably, the incorrect associations made by our strategy are caused by the low absolute value of the judgement operator $J_{i,j}$, and these errors can be avoided by our Algorithm 1 when more than two images are available. For our algorithm, when the DOV is 90°, it is a special case. In this case, our Property 2 is ineffective for the judgment. However, our algorithm still achieves a high association accuracy, which demonstrates the validity of our Property 3 and the necessity of multiple constraints.

TABLE VIII
Time Consumption of PSO-based Approach and AIRLS Algorithm

Number of images	4	8	12	16
AIRLS algorithm	0.008 s	0.021 s	0.060 s	0.122 s
PSO-based approach	1.602 s	4.218 s	10.225 s	16.856 s

TABLE IX
Errors Obtained by Solving Original System Using Different Approaches

		Attitude error (degree)	Size error (m)
PSO-based approach		3.815	0.548
AIRLS algorithm	RI	20.752	1.112
	CEO	4.612	0.787
	CEW	**3.144**	**0.517**

The bold values emphasize that our method achieves lowest estimation errors in experiments under various SNRs.

TABLE X
Errors Obtained by AILS and AIRLS Algorithms

Observation condition		DOV= 2.118°					DOV= 8.822°				
SNR (dB)		-10	-8	-5	0	5	-10	-8	-5	0	5
Attitude error (degree)	AILS	11.778	8.925	5.763	3.448	1.802	9.757	6.076	2.715	1.436	0.692
	AIRLS	11.352	8.483	5.403	3.144	1.785	8.988	5.662	2.558	1.168	0.678
Size error (m)	AILS	6.337	2.611	0.912	0.586	0.250	6.230	2.515	0.728	0.199	0.103
	AIRLS	5.225	2.085	0.808	0.517	0.235	5.587	2.243	0.603	0.175	0.088

TABLE XI
Projection Extraction Errors and Final Weights of Four Images

	1st image	2nd image	3rd image	4th image
Projection extraction errors	0.0386	0.0368	0.0668	0.0431
Final weights	0.4035	0.4072	0.0155	0.1738

4) *Analyze Coarse Estimation Accuracy:* In our coarse estimation, the WS introduces the constraints of multiple properties to ensure estimation accuracy. In the absence of these weights, the coarse estimation is constrained solely by the basic imaging projection model (2), and all the weights in (29) and (30) are set to 1. In Table VI, we compare the errors obtained by the coarse estimation with and without the WS. As shown in Tables IV, V, and VI, the accuracies of the coarse estimation without the WS are quite similar to those of the side-based method [4]. Attributed to the weight assignment based on multiple geometric properties, our coarse estimation also achieves higher estimation accuracies than the existing methods under most observation conditions.

5) *Analyze Performance of AIRLS Algorithm:*

a) *Analyze the convergence:* The convergence of the AIRLS algorithm is proved in Section IV-D3b. To further evaluate the convergence, we conduct Monte-Carlo experiments by randomly initializing the system of equations. Four systems are established using 4, 8, 12, and 16 images, respectively. For each system, 200 Monte-Carlo experiments are conducted. We record the number of iterations required for convergence (NIC), which are present in Table VII. As shown in Table VII, our AIRLS algorithm converges after 4–15 iterations.

b) *Analyze the Computational Complexity:* The objective function (40) for the original system can also be solved using conventional optimization algorithms. In this subsection, we compare the computational complexity and time consumption of our AIRLS algorithm and the PSO-based approach.

Their complexities are listed below:

i) AIRLS algorithm: $O(N_{\text{iter}} \cdot (56N^2 + 62N))$
ii) PSO-based approach: $O(N_{\text{iter}} \cdot N_P \cdot (2N^2 + 10N))$

where N is the number of images, N_{iter} is the number of iterations, and N_P is the number of particles in the PSO algorithm. Both the complexities exhibit a quadratic relationship with the number of images.

In practice, the iteration number N_{iter} for our AIRLS algorithm is usually smaller than 15, whereas the PSO algorithm converges after more than 1800 iterations when N_P is set as 30. Consequently, the total complexity of the PSO-based approach is approximately 130 times that of our AIRLS algorithm. The time consumption obtained by running MATLAB codes on a personal computer with an Intel Core i7 3.40 GHz processor and 16 GB of memory is given in Table VIII. As the number of images increases, time consumption also rises. However, our AIRLS algorithm consistently demonstrates significantly greater efficiency than the PSO-based approach.

c) *Analyze the sensitivity to initial conditions:* Since the objective function (40) is nonconvex, the random initial solutions may lead to convergence at a local optimum. In this section, we employ following three different approaches to initialize the system $\mathbb{S}_{\text{Comp}}$:

1) random initialization (RI);
2) coarse estimation without WS (CEO);
3) coarse estimation with WS (CEW).

The estimation errors obtained by these initialization approaches and the PSO-based approach are given in Table VIII. In this subsection, the DOV is 2.118°, and the SNR is 0 dB. As shown in Table IX, our AIRLS algorithm is sensitive to initial conditions. In our work, we employ CEW to provide a reliable initial solution for the AIRLS algorithm. Due to the reliable initial solution, our AIRLS algorithm achieves higher estimation accuracy than the PSO-based approach, which is robust to local optimums.

d) *Analyze the performance of iterative reweighting:* Without iterative reweighting, the weights for equations are fixed as 1 in the Alternating Iterative Least Square (AILS) algorithm. The estimation errors obtained by the AILS and AIRLS algorithms are listed in Table X. The results indicate that iterative reweighting can slightly improve accuracies under various observation conditions and SNRs.

Then, we analyze whether iterative reweighting can assign smaller weights to abnormal projection components. For each image, its normalized projection extraction error is defined as

$$E_{P,i} = \frac{1}{2}\sum_{k=1}^{2}\frac{\left\|\overrightarrow{\mathbf{Ip}}_{t,k,i} - \overrightarrow{\mathbf{Ip}}_{e,k,i}\right\|_2}{\left\|\overrightarrow{\mathbf{Ip}}_{t,k,i}\right\|_2} \tag{55}$$

where $\overrightarrow{\mathbf{Ip}}_{t,k,i}$ and $\overrightarrow{\mathbf{Ip}}_{e,k,i}$ $(k = 1, 2)$ are, respectively, the theoretical and extracted side projections in the ith image. The final weight for the ith image is calculated as

$$\omega_{f,i} = \omega_{2(i-1)+1} + \omega_{2(i-1)+2} + \omega_{2N+1+i}$$

TABLE XII
Mean Computation Time of Different Methods

Number of images	4	8	12	16
Side-based method	0.823 s	2.018 s	3.788 s	5.889 s
PCA-based method	3.215 s	6.085 s	10.384 s	12.755 s
QPSV-based method	4.728 s	8.855 s	14.329 s	17.668 s
Coarse estimation in our method	0.002 s	0.003 s	0.005 s	0.009 s
Our method	0.011 s	0.024 s	0.065 s	0.132 s

TABLE XIII
Estimation Errors Obtained With Limited Optical Observations Under Different SNRs (DOV $=3.029^\circ$)

	Attitude estimation errors (degree)					Size estimation errors (m)				
SNR (dB)	-10	-8	-5	0	5	-10	-8	-5	0	5
Side-based method	18.557	10.583	9.902	6.826	5.166	6.077	2.485	1.212	1.025	0.856
PCA-based method	20.248	14.377	11.888	8.791	6.055	8.305	4.175	2.061	1.458	1.095
Coarse estimation in our method	11.442	8.024	4.093	3.293	2.118	4.270	1.492	0.672	0.452	0.402
Our method	**7.566**	**5.827**	**2.915**	**2.325**	**1.769**	**1.867**	**0.836**	**0.524**	**0.403**	**0.378**

The bold values emphasize that our method achieves lowest estimation errors in experiments under various SNRs.

$$\omega_{f,i} = \omega_{f,i} / \sum_{i=1}^{N} \omega_{f,i} \tag{56}$$

where $\omega_{2(i-1)+1}$, $\omega_{2(i-1)+2}$, and ω_{2N+1+i} are elements of the weight vector output by the AIRLS algorithm.

In Table XI, we list projection extraction errors and final weights of four images (DOV $=2.118^\circ$, SNR $=0$ dB). The smallest weight is assigned to the third image, whose projection extraction error is obviously larger, indicating the effectiveness of iterative reweighting.

6) *Analyze Computation Time:* Notably, the optimizations in the comparative methods are also nonconvex [4], [8], [10]. These optimizations are usually solved by the PSO algorithm. In our experiments, we set the particle number as 30, and the PSO algorithm converges after more than 1200 iterations. In comparison, our method avoids the time-consuming optimization process. Although the subsystems are solved with an iterative strategy, our method quickly converges after 4–15 iterations. Table XII presents the computation time of different methods. Our method takes much less computation time than the existing methods.

B. Experiments on Optical Images

In this part, optical images of the Aura satellite are simulated by the ray tracing algorithm [34]. The optoelectronic telescope is located in Hangzhou, and the target attitude is consistent with that in experiments on radar images. Under this experimental condition, its observation angles in the RTN system are the same as those of the radar. The image resolution is 0.05 meters, and the imaging interval is 10 s. Due to different imaging mechanisms, the DOVs of the optical images are different from those of the radar images, and the variation curves are given in Fig. 13(a). Some simulated images are shown in Fig. 13(b)–(d).

The attitude and size estimation errors obtained with limited observations (30 s, 4 images, DOV $=3.029^\circ$) are shown in Table XIII. The PCA-based method achieves the worst estimation accuracy in experiments on optical images.

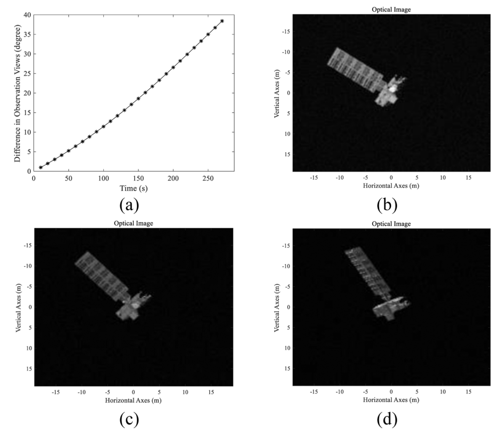

Fig. 13. Simulated optical images of the Aura satellite. (a) DOVs of the optical image sequence. (b)–(c) Simulated optical images.

Compared with the side-based method, the estimation errors of our method are 4.413° and 0.873 m lower on average, which indicates that our geometric properties remain valid in optical imaging.

C. Some Discussions

The experimental analysis above demonstrates that our method can accurately estimate attitude and size of the satellite under limited observation conditions. To promote its practical application, some issues remain to be addressed.

In most methods, the estimation process is divided into two steps: projection feature extraction and cross-view processing. Our work improves estimation accuracy by leveraging the geometric properties during the cross-view processing. The estimation accuracy can also be improved by modifying the projection feature extraction algorithms. For solar panels, the projection feature extraction is usually realized by image processing operations, such as satellite component segmentation and parallelogram parameter estimation. The semantic segmentation algorithm based on the Pix2pixGAN network has been widely utilized for satellite component segmentation [7]. However, its noise immunity and generalization should be further enhanced. Differences in satellite structure, SNR, and data source can lead to poor segmentation results. It is expensive to label images of various satellites with different attitudes for training. To decrease its reliance on training data, incorporating the prior of geometric shapes into the semantic segmentation networks can be a potential approach. The boundary-based algorithms are typically employed in parallelogram parameter estimation [7]. However, the boundary information is noise-sensitive. There is an urgent need to improve its noise-robustness.

Although we summarize and derive the geometric properties in the scenario of observing three-axis stabilized satellites using ground-based sensors, these properties are also applicable to a broader range of scenarios. For space-borne observation, its imaging projection model should be established according to its specific imaging mechanism [37], utilizing both sensor GPS data and target tracking data. For rotating satellites, their imaging projection models should be determined by dynamic analysis methods first [15], [16], [17], [18].

In addition, our method is primarily applicable to common satellites with one solar panel or multiple parallel solar panels. When multiple solar panels of a satellite are unparallel, it is necessary to associate the same solar panel among different images before associating the sides. Although our properties still hold true, the association algorithm should be specifically redesigned for such cases.

VI. CONCLUSION

In this article, we propose a satellite attitude and size estimation method based on imaging geometry analysis to improve estimation accuracies, especially when view differences are small. Based on the proven geometric properties about space observation, an over-determined system of equations is established to estimate two adjacent side vectors of the rectangular solar panel from their image projections. To solve this complex system, a coarse estimation is implemented in advance to decouple the system into linear subsystems, and these subsystems are alternatively and iteratively solved for the final delicate solution. The experimental results validate the effectiveness of our methods on multimode data with different SNRs. In comparison to existing methods, our method achieves higher estimation accuracy and costs less computation time, making it more suitable for practical application.

APPENDIX I
PROOF FOR FIVE PROPERTIES

Calculate the projection components $\vec{\mathbf{p}}_{k,i}$ $(k = 1, 2)$

$$\vec{\mathbf{p}}_{1,i} = \vec{\mathbf{a}} - \left(\vec{\mathbf{a}}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i = a\left(\left(y_i^2+z_i^2\right)\cdot\left(\frac{\vec{\mathbf{a}}}{a}\right) - x_iy_i\cdot\left(\frac{\vec{\mathbf{b}}}{b}\right) - x_iz_i\cdot\vec{\mathbf{z}}\right) \tag{57}$$

$$\vec{\mathbf{p}}_{2,i} = \vec{\mathbf{b}} - \left(\vec{\mathbf{b}}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i = b\left(\left(x_i^2+z_i^2\right)\cdot\left(\frac{\vec{\mathbf{b}}}{b}\right) - x_iy_i\cdot\left(\frac{\vec{\mathbf{a}}}{a}\right) - y_iz_i\cdot\vec{\mathbf{z}}\right). \tag{58}$$

Calculate the normal vectors $\vec{\mathbf{np}}_{k,i}$ of the planes $Pl_{k,i}$

$$\vec{\mathbf{np}}_{1,i} = \frac{\vec{\mathbf{n}}_i\otimes\vec{\mathbf{p}}_{1,i}}{\left\|\vec{\mathbf{n}}_i\otimes\vec{\mathbf{p}}_{1,i}\right\|_2} = \frac{1}{\sqrt{y_i^2+z_i^2}}\cdot\left(z_i\cdot\left(\frac{\vec{\mathbf{b}}}{b}\right) - y_i\cdot\vec{\mathbf{z}}\right) \tag{59}$$

$$\vec{\mathbf{np}}_{2,i} = \frac{\vec{\mathbf{n}}_i\otimes\vec{\mathbf{p}}_{2,i}}{\left\|\vec{\mathbf{n}}_i\otimes\vec{\mathbf{p}}_{2,i}\right\|_2} = \frac{1}{\sqrt{x_i^2+z_i^2}}\cdot\left(z_i\cdot\left(\frac{\vec{\mathbf{a}}}{a}\right) - x_i\cdot\vec{\mathbf{z}}\right). \tag{60}$$

Calculate the intersection lines $\vec{\mathbf{l}}_{k,i,j}$ $(k = 1, 2, 3, 4)$

$$\vec{\mathbf{l}}_{1,i,j} = \frac{\vec{\mathbf{np}}_{1,i}\otimes\vec{\mathbf{np}}_{1,j}}{\left\|\vec{\mathbf{np}}_{1,i}\otimes\vec{\mathbf{np}}_{1,j}\right\|_2} = \frac{\vec{\mathbf{a}}}{a} \tag{61}$$

$$\vec{\mathbf{l}}_{2,i,j} = \frac{\vec{\mathbf{np}}_{2,i}\otimes\vec{\mathbf{np}}_{2,j}}{\left\|\vec{\mathbf{np}}_{2,i}\otimes\vec{\mathbf{np}}_{2,j}\right\|_2} = \frac{\vec{\mathbf{b}}}{b} \tag{62}$$

$$\vec{\mathbf{l}}_{3,i,j} = \frac{\vec{\mathbf{np}}_{1,i}\otimes\vec{\mathbf{np}}_{2,j}}{\left\|\vec{\mathbf{np}}_{1,i}\otimes\vec{\mathbf{np}}_{2,j}\right\|_2} = \frac{z_ix_j\cdot\frac{\vec{\mathbf{a}}}{a} + y_iz_j\cdot\frac{\vec{\mathbf{b}}}{b} + z_iz_j\cdot\vec{\mathbf{z}}}{\xi_{l_3}} \tag{63}$$

$$\vec{\mathbf{l}}_{4,i,j} = \frac{\vec{\mathbf{np}}_{2,i}\otimes\vec{\mathbf{np}}_{1,j}}{\left\|\vec{\mathbf{np}}_{2,i}\otimes\vec{\mathbf{np}}_{1,j}\right\|_2} = \frac{x_iz_j\cdot\frac{\vec{\mathbf{a}}}{a} + z_iy_j\cdot\frac{\vec{\mathbf{b}}}{b} + z_iz_j\cdot\vec{\mathbf{z}}}{\xi_{l_4}} \tag{64}$$

where $\xi_{l_3} = \sqrt{z_i^2x_j^2 + y_i^2z_j^2 + z_i^2z_j^2}$, $\xi_{l_4} = \sqrt{x_i^2z_j^2 + z_i^2y_j^2 + z_i^2z_j^2}$.

Calculate the cross product $\vec{\mathbf{n}}_{z,i,j}$

$$\vec{\mathbf{n}}_{z,i,j} = \frac{\vec{\mathbf{n}}_i\otimes\vec{\mathbf{n}}_j}{\left\|\vec{\mathbf{n}}_i\otimes\vec{\mathbf{n}}_j\right\|_2} = \frac{\xi_b\cdot\frac{\vec{\mathbf{a}}}{a} + \xi_a\cdot\frac{\vec{\mathbf{b}}}{b} + \xi_z\cdot\vec{\mathbf{z}}}{\sqrt{\xi_a^2+\xi_b^2+\xi_z^2}} \tag{65}$$

where $\xi_a = x_jz_i - x_iz_j$, $\xi_b = y_iz_j - y_jz_i$, and $\xi_z = x_iy_j - x_jy_i$.

1) Proof for Property 1
According to (61) and (62), $\vec{\mathbf{l}}_{1,i,j}$ is parallel to $\vec{\mathbf{a}}$, and $\vec{\mathbf{l}}_{2,i,j}$ is parallel to $\vec{\mathbf{b}}$.
2) Proof for Property 2
2-1) Proof for $\vec{\mathbf{l}}_{1,i,j}$ is always perpendicular to $\vec{\mathbf{l}}_{2,i,j}$.
Since two sides of a rectangle are orthogonal, $\vec{\mathbf{a}}^T\vec{\mathbf{b}} = 0$.
According to (61) and (62): $\vec{\mathbf{l}}_{1,i,j}^T\vec{\mathbf{l}}_{2,i,j} = \frac{\vec{\mathbf{a}}^T\vec{\mathbf{b}}}{ab} = 0$.
2-2) Proof for $\vec{\mathbf{l}}_{3,i,j}$ is not perpendicular to $\vec{\mathbf{l}}_{4,i,j}$ when $\vec{\mathbf{n}}_i^T\vec{\mathbf{n}}_j \neq 0$.
According to (63) and (64)

$$\vec{\mathbf{l}}_{3,i,j}^T\vec{\mathbf{l}}_{4,i,j} = \frac{z_iz_j\cdot\left(x_ix_j+y_iy_j+z_iz_j\right)}{\xi_{l_3}\xi_{l_4}} = \frac{z_iz_j}{\xi_{l_3}\xi_{l_4}}\cdot\vec{\mathbf{n}}_i^T\vec{\mathbf{n}}_j.$$

Since $\vec{\mathbf{n}}_i^T\vec{\mathbf{n}}_j \neq 0$, $z_{i/j} \neq 0$, and $\xi_{l_3}, \xi_{l_4} > 0$, $\vec{\mathbf{l}}_{3,i,j}^T\vec{\mathbf{l}}_{4,i,j} \neq 0$.

3) Proof for Property 3

3-1) Proof for $PE_{3,1,i,j}$ and $PE_{3,2,i,j}$ are always valid. According to (65), we can obtain $\vec{\mathbf{n}}_i^T\vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{n}}_j^T\vec{\mathbf{n}}_{z,i,j} = 0$.

According to (57) and (58), we can obtain (66)–(69)

$$\begin{aligned}\vec{\mathbf{p}}_{1,i}^T\vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{p}}_{1,i}^T\vec{\mathbf{n}}_{z,i,j} &= \left(\vec{\mathbf{a}} - \left(\vec{\mathbf{a}}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i\right)^T\vec{\mathbf{n}}_{z,i,j}\\ &= \vec{\mathbf{a}}^T\vec{\mathbf{n}}_{z,i,j} - \left(\vec{\mathbf{a}}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i^T\vec{\mathbf{n}}_{z,i,j}\\ &= \vec{\mathbf{a}}^T\vec{\mathbf{n}}_{z,i,j}\end{aligned} \tag{66}$$

$$\vec{\mathbf{p}}_{1,j}^T\vec{\mathbf{n}}_{z,i,j} = \left(\vec{\mathbf{a}} - \left(\vec{\mathbf{a}}^T\vec{\mathbf{n}}_j\right)\cdot\vec{\mathbf{n}}_j\right)^T\vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{a}}^T\vec{\mathbf{n}}_{z,i,j} \tag{67}$$

$$\vec{\mathbf{p}}_{2,i}^T\vec{\mathbf{n}}_{z,i,j} = \left(\vec{\mathbf{b}} - \left(\vec{\mathbf{b}}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i\right)^T\vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{b}}^T\vec{\mathbf{n}}_{z,i,j} \tag{68}$$

$$\vec{\mathbf{p}}_{2,j}^T\vec{\mathbf{n}}_{z,i,j} = \left(\vec{\mathbf{b}} - \left(\vec{\mathbf{b}}^T\vec{\mathbf{n}}_j\right)\cdot\vec{\mathbf{n}}_j\right)^T\vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{b}}^T\vec{\mathbf{n}}_{z,i,j}. \tag{69}$$

Therefore, $\vec{\mathbf{p}}_{1,i}^T\vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{p}}_{1,j}^T\vec{\mathbf{n}}_{z,i,j}$ $(PE_{3,1,i,j})$, and $\vec{\mathbf{p}}_{2,i}^T\vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{p}}_{2,j}^T\vec{\mathbf{n}}_{z,i,j}$. $(PE_{3,2,i,j})$.

3-2) Proof for $PE_{3,3,i,j}$ and $PE_{3,4,i,j}$ are not valid when $\xi_a\cdot b \neq \xi_b\cdot a$.

According to (65) and (66), $\vec{\mathbf{p}}_{1,i}^T\vec{\mathbf{n}}_{z,i,j} = \frac{a\cdot\xi_b}{\sqrt{\xi_a^2+\xi_b^2+\xi_z^2}}$.

According to (65) and (69), $\vec{\mathbf{p}}_{2,j}^T\vec{\mathbf{n}}_{z,i,j} = \frac{b\cdot\xi_a}{\sqrt{\xi_a^2+\xi_b^2+\xi_z^2}}$.

Since $\xi_a\cdot b \neq \xi_b\cdot a$, $\frac{a\cdot\xi_b}{\sqrt{\xi_a^2+\xi_b^2+\xi_z^2}} \neq \frac{b\cdot\xi_a}{\sqrt{\xi_a^2+\xi_b^2+\xi_z^2}}$.

Therefore, $\vec{\mathbf{p}}_{1,i}^T\vec{\mathbf{n}}_{z,i,j} \neq \vec{\mathbf{p}}_{2,j}^T\vec{\mathbf{n}}_{z,i,j}$ ($PE_{3,3,i,j}$ is not valid). Similarly, according to (65), (67), and (68), we can prove that $PE_{3,4,i,j}$ is not valid.

4) Proof for Property 4

Considering that the parallelogram vertexes in the ith imaging plane are $\vec{\mathbf{v}}_{p,1,i} = (\vec{\mathbf{p}}_{1,i} + \vec{\mathbf{p}}_{2,i})/2$, $\vec{\mathbf{v}}_{p,2,i} = (\vec{\mathbf{p}}_{1,i} - \vec{\mathbf{p}}_{2,i})/2$, $\vec{\mathbf{v}}_{p,3,i} = -(\vec{\mathbf{p}}_{1,i} + \vec{\mathbf{p}}_{2,i})/2$, and $\vec{\mathbf{v}}_{p,4,i} = (\vec{\mathbf{p}}_{2,i} - \vec{\mathbf{p}}_{1,i})/2$, vertexes of the first corresponding rectangle are: $\vec{\mathbf{v}}_{r,1,1} = (\vec{\mathbf{a}} + \vec{\mathbf{b}})/2$, $\vec{\mathbf{v}}_{r,2,1} = (\vec{\mathbf{a}} - \vec{\mathbf{b}})/2$, $\vec{\mathbf{v}}_{r,3,1} = -(\vec{\mathbf{a}} + \vec{\mathbf{b}})/2$, $\vec{\mathbf{v}}_{r,4,1} = (\vec{\mathbf{b}} - \vec{\mathbf{a}})/2$.

Vertexes of the second corresponding rectangle are: $\vec{\mathbf{v}}_{r,1,2} = \xi_1\cdot\vec{\mathbf{l}}_{3,i,j} - \xi_2\cdot\vec{\mathbf{l}}_{4,i,j}$, $\vec{\mathbf{v}}_{r,2,2} = \xi_1\cdot\vec{\mathbf{l}}_{3,i,j} + \xi_2\cdot\vec{\mathbf{l}}_{4,i,j}$, $\vec{\mathbf{v}}_{r,3,2} = \xi_2\cdot\vec{\mathbf{l}}_{4,i,j} - \xi_1\cdot\vec{\mathbf{l}}_{3,i,j}$, and $\vec{\mathbf{v}}_{r,4,2} = -\xi_1\cdot\vec{\mathbf{l}}_{3,i,j} - \xi_2\cdot\vec{\mathbf{l}}_{4,i,j}$

where $\xi_1 = \xi_A\xi_{l_3}/2$, $\xi_2 = \xi_B\xi_{l_4}/2$, $\xi_A = a/\xi_a$, $\xi_B = b/\xi_b$.

They can be proved by analyzing the side orthogonality and vertex projections. The details are as follows.

4.1) Proof for the first planar rectangle

Its two adjacent sides are $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$, and they are orthogonal.

The projection of $\vec{\mathbf{v}}_{r,1}$ is calculated as<?TeX

$$\begin{aligned}\vec{\mathbf{v}'}_{p,1,i} &= \vec{\mathbf{v}}_{r,1,1} - \left(\vec{\mathbf{v}}_{r,1}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i\\ &= \frac{1}{2}\left(\vec{\mathbf{a}} - \left(\vec{\mathbf{a}}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i + \vec{\mathbf{b}} - \left(\vec{\mathbf{b}}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i\right)\\ &= \frac{1}{2}\left(\vec{\mathbf{p}}_{1,i} + \vec{\mathbf{p}}_{2,i}\right) = \vec{\mathbf{v}}_{p,1,i}.\end{aligned} \tag{70}$$

Similarly, we can get that $\vec{\mathbf{v}'}_{p,1,j} = \vec{\mathbf{v}}_{p,1,j}$, $\vec{\mathbf{v}'}_{p,2,i/j} = \vec{\mathbf{v}}_{p,2,i/j}$, $\vec{\mathbf{v}'}_{p,3,i/j} = \vec{\mathbf{v}}_{p,3,i/j}$, and $\vec{\mathbf{v}'}_{p,4,i/j} = \vec{\mathbf{v}}_{p,4,i/j}$. Therefore, its projection components on the ith and jth imaging plane are always $\vec{\mathbf{p}}_{k,i}$ and $\vec{\mathbf{p}}_{k,j}$ $(k = 1, 2)$.

4.2) Proof for the second planar rectangle

Its two adjacent sides are $\vec{\mathbf{v}}_{r,1,2} - \vec{\mathbf{v}}_{r,4,2} = 2\xi_1\cdot\vec{\mathbf{l}}_{3,i,j}$ and $\vec{\mathbf{v}}_{r,2,2} - \vec{\mathbf{v}}_{r,1,2} = 2\xi_2\cdot\vec{\mathbf{l}}_{4,i,j}$. When $\vec{\mathbf{n}}_i^T\vec{\mathbf{n}}_j = 0$, $\vec{\mathbf{l}}_{3,i,j}^T\vec{\mathbf{l}}_{4,i,j} = 0$. Therefore, the sides are orthogonal.

Calculate the projections of $\vec{\mathbf{l}}_{3,i,j}$ and $\vec{\mathbf{l}}_{4,i,j}$

$$\begin{aligned}\vec{\mathbf{p}}_{l3,i} &= \vec{\mathbf{l}}_{3,i,j} - \left(\vec{\mathbf{l}}_{3,i,j}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i\\ &= \frac{\xi_a}{\xi_{l_3}}\cdot\left(\left(y_i^2 + z_i^2\right)\cdot\left(\frac{\vec{\mathbf{a}}}{a}\right) - x_iy_i\cdot\left(\frac{\vec{\mathbf{b}}}{b}\right)\right.\\ &\quad\left.-x_iz_i\cdot\vec{\mathbf{z}}\right)\\ &= \frac{\xi_a\cdot\vec{\mathbf{p}}_{1,i}}{a\xi_{l_3}} = \frac{\vec{\mathbf{p}}_{1,i}}{\xi_A\xi_{l_3}}\end{aligned} \tag{71}$$

$$\vec{\mathbf{p}}_{l3,j} = \vec{\mathbf{l}}_{3,i,j} - \left(\vec{\mathbf{l}}_{3,i,j}^T\vec{\mathbf{n}}_j\right)\cdot\vec{\mathbf{n}}_j = \frac{\vec{\mathbf{p}}_{2,j}}{\xi_B\xi_{l_3}} \tag{72}$$

$$\vec{\mathbf{p}}_{l4,i} = \vec{\mathbf{l}}_{4,i,j} - \left(\vec{\mathbf{l}}_{4,i,j}^T\vec{\mathbf{n}}_i\right)\cdot\vec{\mathbf{n}}_i = -\frac{\vec{\mathbf{p}}_{2,i}}{\xi_B\xi_{l_4}} \tag{73}$$

$$\vec{\mathbf{p}}_{l4,j} = \vec{\mathbf{l}}_{4,i,j} - \left(\vec{\mathbf{l}}_{4,i,j}^T\vec{\mathbf{n}}_j\right)\cdot\vec{\mathbf{n}}_j = -\frac{\vec{\mathbf{p}}_{1,j}}{\xi_A\xi_{l_4}}. \tag{74}$$

Then, the projections of $\vec{\mathbf{v}}_{r,1,2}$ are as follows:

$$\begin{aligned}\vec{\mathbf{v}'}_{p,1,i} &= \frac{\xi_A\xi_{l_3}\cdot\vec{\mathbf{p}}_{l3,i}}{2} - \frac{\xi_B\xi_{l_4}\cdot\vec{\mathbf{p}}_{l4,i}}{2}\\ &= \frac{1}{2}\left(\vec{\mathbf{p}}_{1,i} + \vec{\mathbf{p}}_{2,i}\right) = \vec{\mathbf{v}}_{p,1,i}\end{aligned} \tag{75}$$

$$\begin{aligned}\vec{\mathbf{v}'}_{p,1,j} &= \frac{\xi_A\xi_{l_3}\cdot\vec{\mathbf{p}}_{l3,j}}{2} - \frac{\xi_B\xi_{l_4}\cdot\vec{\mathbf{p}}_{l4,j}}{2}\\ &= \frac{1}{2}\left(\frac{\xi_A\cdot\vec{\mathbf{p}}_{2,j}}{\xi_B} + \frac{\xi_B\cdot\vec{\mathbf{p}}_{1,j}}{\xi_A}\right).\end{aligned} \tag{76}$$

When $\xi_a\cdot b = \xi_b\cdot a$, $\xi_A = \xi_B$. Therefore, we can obtain $\vec{\mathbf{v}'}_{p,1,j} = (\vec{\mathbf{p}}_{1,j} + \vec{\mathbf{p}}_{2,j})/2 = \vec{\mathbf{v}}_{p,1,j}$. Similarly, we can get that $\vec{\mathbf{v}'}_{p,1,j} = \vec{\mathbf{v}}_{p,1,j}$, $\vec{\mathbf{v}'}_{p,1,j} = \vec{\mathbf{v}}_{p,1,j}$, and $\vec{\mathbf{v}'}_{p,1,j} = \vec{\mathbf{v}}_{p,1,j}$. Therefore, its projection components on the ith and jth imaging plane are $\vec{\mathbf{p}}_{k,i}$ and $\vec{\mathbf{p}}_{k,j}$ $(k = 1, 2)$ when $\xi_a\cdot b = \xi_b\cdot a$ and $\vec{\mathbf{n}}_i^T\vec{\mathbf{n}}_j = 0$.

5) Proof for Property 5

Calculate $\frac{(\vec{\mathbf{l}}_{1,i,j}^T\vec{\mathbf{l}}_{3,i,j})\cdot(\vec{\mathbf{n}}_i^T\vec{\mathbf{l}}_{2,i,j})}{\vec{\mathbf{n}}_i^T\vec{\mathbf{z}}}$

$$\frac{\left(\vec{\mathbf{l}}_{1,i,j}^T\vec{\mathbf{l}}_{3,i,j}\right)\cdot\left(\vec{\mathbf{n}}_i^T\vec{\mathbf{l}}_{2,i,j}\right)}{\vec{\mathbf{n}}_i^T\vec{\mathbf{z}}} = \frac{\left(z_ix_j\right)\cdot y_i}{\xi_{l_3}\cdot z_i} = \frac{x_jy_i}{\xi_{l_3}}. \tag{77}$$

Calculate $\frac{(\vec{\mathbf{l}}_{2,i,j}^T\vec{\mathbf{l}}_{3,i,j})\cdot(\vec{\mathbf{n}}_j^T\vec{\mathbf{l}}_{1,i,j})}{\vec{\mathbf{n}}_j^T\vec{\mathbf{z}}}$

$$\frac{\left(\vec{\mathbf{l}}_{2,i,j}^T\vec{\mathbf{l}}_{3,i,j}\right)\cdot\left(\vec{\mathbf{n}}_{2,i,j}^T\vec{\mathbf{l}}_{1,i,j}\right)}{\vec{\mathbf{n}}_j^T\vec{\mathbf{z}}} = \frac{\left(y_iz_j\right)\cdot x_j}{\xi_{l_3}\cdot z_j} = \frac{x_jy_i}{\xi_{l_3}}. \tag{78}$$

Therefore, $PE_{5,1,i,j}$ is always valid. Similarly, we can prove $PE_{5,2,i,j}$.

APPENDIX II
CALCULATION OF NORMAL VECTORS AND INTERSECTION LINES

In this section, we provide the details of calculating $\vec{\mathbf{n}}_i$, $\vec{\mathbf{n}}_j$, $\vec{\mathbf{n}}_{z,i,j}$, $\vec{\mathbf{l}}_{1,i,j}$, $\vec{\mathbf{l}}_{2,i,j}$, $\vec{\mathbf{l}}_{3,i,j}$, $\vec{\mathbf{l}}_{4,i,j}$, and $\vec{\mathbf{z}}$ from the imaging projection matrices $\mathbf{H}_i$, $\mathbf{H}_j$ and projection components $\vec{\mathbf{p}}_{k,i}$ $\vec{\mathbf{p}}_{k,j}$ $(k = 1, 2)$.

For the ith image, the normal vector $\vec{\mathbf{n}}_i$ of its imaging plane is calculated as

$$\vec{\mathbf{n}}_i = \frac{\mathbf{H}_i\,[1] \otimes \mathbf{H}_i\,[2]}{\|\mathbf{H}_i\,[1] \otimes \mathbf{H}_i\,[2]\|_2} \tag{79}$$

where $\mathbf{H}_i[1]$ and $\mathbf{H}_i[2]$ are, respectively, the first and second rows of the imaging projection matrix $\mathbf{H}_i$. Similarly, we can obtain $\vec{\mathbf{n}}_j$ and $\vec{\mathbf{n}}_{z,i,j} = \vec{\mathbf{n}}_i \otimes \vec{\mathbf{n}}_j$.

Then, we can determine the plane $Pl_{k,i}$ by $\vec{\mathbf{n}}_i$ and the projection component $\vec{\mathbf{p}}_{k,i}$. Its normal vectors $\overrightarrow{\mathbf{np}}_{k,i}$ $(k = 1, 2)$ is calculated as

$$\overrightarrow{\mathbf{np}}_{k,i} = \frac{\vec{\mathbf{n}}_i \otimes \vec{\mathbf{p}}_{k,i}}{\left\|\vec{\mathbf{n}}_i \otimes \vec{\mathbf{p}}_{k,i}\right\|_2}. \tag{80}$$

Similarly, we can obtain $\overrightarrow{\mathbf{np}}_{k,j}$.

Finally, the intersection line $\vec{\mathbf{l}}_{1,i,j}$ between $Pl_{1,i}$ and $Pl_{1,j}$ is calculated as

$$\vec{\mathbf{l}}_{1,i,j} = \frac{\overrightarrow{\mathbf{np}}_{1,i} \otimes \overrightarrow{\mathbf{np}}_{1,j}}{\left\|\overrightarrow{\mathbf{np}}_{1,i} \otimes \overrightarrow{\mathbf{np}}_{1,j}\right\|_2}. \tag{81}$$

Similarly, we can obtain the intersection line $\vec{\mathbf{l}}_{2,i,j}$ between $Pl_{2,i}$ and $Pl_{2,j}$, the intersection line $\vec{\mathbf{l}}_{3,i,j}$ between $Pl_{1,i}$ and $Pl_{2,j}$, and the intersection line $\vec{\mathbf{l}}_{4,i,j}$ between $Pl_{2,i}$ and $Pl_{1,j}$. Then, $\vec{\mathbf{z}} = \vec{\mathbf{l}}_{1,i,j} \otimes \vec{\mathbf{l}}_{2,i,j}$.

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